What Prices Cannot Tell a Market Supervisor: The Cost of Identifying Trader Groups and the Value of Class Disclosure

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Working paper: 5 November 2026

Abstract

What can prices reveal about trader groups, and when is additional disclosure worth its cost? In a linear feedback market, prices identify a merged distribution of trading responses; equal-memory groups remain inseparable, while nearby distinct groups can require sharply increasing data. We translate identification difficulty into sample, execution and capital requirements under explicit observation contracts. A two-group supervisory example separates an aggregate prudential constraint from a targeted halt decision: nearly indistinguishable price responses can imply different feasible actions. General-population extensions require a separately verified dominant-group gap. We replace an unrestricted strategic-response claim with a bounded-response condition and connect the example to information fees, intervention losses and decision errors. A reproducible synthetic accounting study exhibits both positive and negative values of paid information and failures under bias or underestimated responses. Existing position-report studies are descriptive diagnostics; they do not identify learning memory, validate the intervention model, or establish welfare gains. Proof review, the observation-to-market bridge and the precise incremental contribution remain explicit research obligations.

AI-use disclosure. Claude, ChatGPT/Codex and Gemini provided material assistance with research development, code, numerical checks, manuscript drafting and internal critique.

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ChatGPT/Codex additionally developed and checked Lean proofs in this revision. Human authors remain responsible for verifying the content and its scholarly contribution before submission; AI checks do not constitute independent review.

Keywords: algorithmic trading, identification, trader heterogeneity, market microstructure, system identification, minimax sample complexity, value of disclosure, audit trails.

JEL: C14, C32, C58, D83, G14, G18.

Code and data. All code, pre-specified test specifications and result files are public at https://github.com/Pratikn03/Fianince.

Reproducibility and verification. The appendix maps historical tables and figures to generating code and result files. Existing empirical results are retained with their original specifications, failures and limitations; this revision does not independently rerun every historical study. "Pre-specified" denotes the authors' dated specification and repository record, not independent registration. The new accounting study reproduces six archived response rows and records both the normal-quantile and conservative gates, including adverse configurations. The pinned Lean build and strict axiom audit verify typed declarations; the exact supervisory witness, conservative Gaussian event bound and acquisition algebra have explicit coverage below. Historical name mappings or declaration counts do not establish full manuscript equivalence. The stationary IV central limit argument, external approximation results, general-K supervisory construction and economic observation assumptions retain their stated limits. Code and receipts accompany this working paper; independent proof review remains a separate task.

1 Introduction

Markets are increasingly run by algorithms. Trend-following funds, market-making programs, execution algorithms and, more recently, AI trading agents all read recent prices, decide what to trade, and by trading move the prices they read. When something goes wrong, a sharp sell-off, a sudden loss of liquidity, a crowded trade unwinding, the first questions supervisors and risk managers ask are who was trading and how. Which group of traders drove the move? How quickly do they react, and how long do they remember what prices did? Could the same group do it again?

This paper asks a question that supervisors now face when they decide which trading records to require: which questions about who drives prices can public data answer, which can they answer only at a prohibitive cost, and what is it worth to be told the rest?

The question matters now. Trading is passing to learning algorithms and AI agents (Colliard et al., 2022; Dou et al., 2025), and regulators have begun to require firms to say which algorithm sent each order. Under MiFID II, EU trading venues must keep order records that identify the person or algorithm responsible for each investment and execution decision (European Commission, 2017; ESMA, 2016). The US Consolidated Audit Trail (CAT) is designed to record each order in US-listed stocks and options, and the account behind it, across exchanges and broker-dealers; its revised 2026 budget, published in March 2026, is about $147 million (SEC, 2012; CAT LLC, 2026). Such systems serve many purposes, among them enforcement, rebuilding market events after the fact, and surveillance. One question they raise, and the one we study, is how much of the information about who drives price moves could instead be learned from public data, and at what cost.

The question. A supervisor wants to know which group of traders drives prices. What can public prices tell it, what does it cost to learn the rest, and what are trader-class reports worth?

The answer.

1. *Prices see the crowd, not the groups.* They reveal one combined summary of all the groups. Two groups with the same memory look exactly like one larger group, however much price data is collected (Theorem 3.1).
2. *So supervision splits in two.* A market-wide measure depends on an aggregate response, subject to the observation and tail conditions. A targeted decision can require more information: two markets can look almost identical in prices while halting the largest group meets the prudential ceiling in one and not in the other (Proposition 6.4, machine-checked for two groups).
3. *Learning the groups from prices is very expensive.* The data needed grow as $(\Delta\eta)^{-(4K-2)}$ when the memories of K groups are $\Delta\eta$ apart: halving the gap multiplies the data by 64 for two groups and by 1,024 for three (Theorems 5.1 and 5.2). If memory is a general mixture, accuracy improves only with the logarithm of the data (Theorem S17.2).
4. *Class records change the information requirement.* When each class's flow is reported separately, the problem stays well conditioned however similar the groups are (Proposition 6.1). In 20 futures markets, the CFTC's weekly class report pins down the persistence of speculative positions 9 to 71 times more precisely than prices; a hypothetical common-root-T conversion gives about 1,350 at the median, but does not identify an equivalent sample requirement (Figure 1).
5. *What we do not show.* That fitted persistence is the traders' learning memory: the weekly reports constrain how persistent class positions are, which can also reflect turnover, reporting aggregation or inventory retention. Of 23 recorded outcomes, 11 failed (10 of the 19 primary ones; Table S29). One possible reason, which we have not tested, is that a reporting class mixes strategies that respond to prices with opposite signs.

We keep three statements apart throughout: what cannot be learned at all, what can be learned but only at high cost, and where estimates become unstable.

Precise comparison with existing identification research. Clustered exponential recovery already explains ill-conditioning and separation-dependent rates (Batenkov et al., 2021; Katz et al., 2023); those exponents are not claimed as a general new discovery. Data informativity already distinguishes information sufficient for a control task from unique system identification (van Waarde et al., 2020); the broad distinction is likewise not new. Our candidate contribution is their explicit combination in this market feedback model:

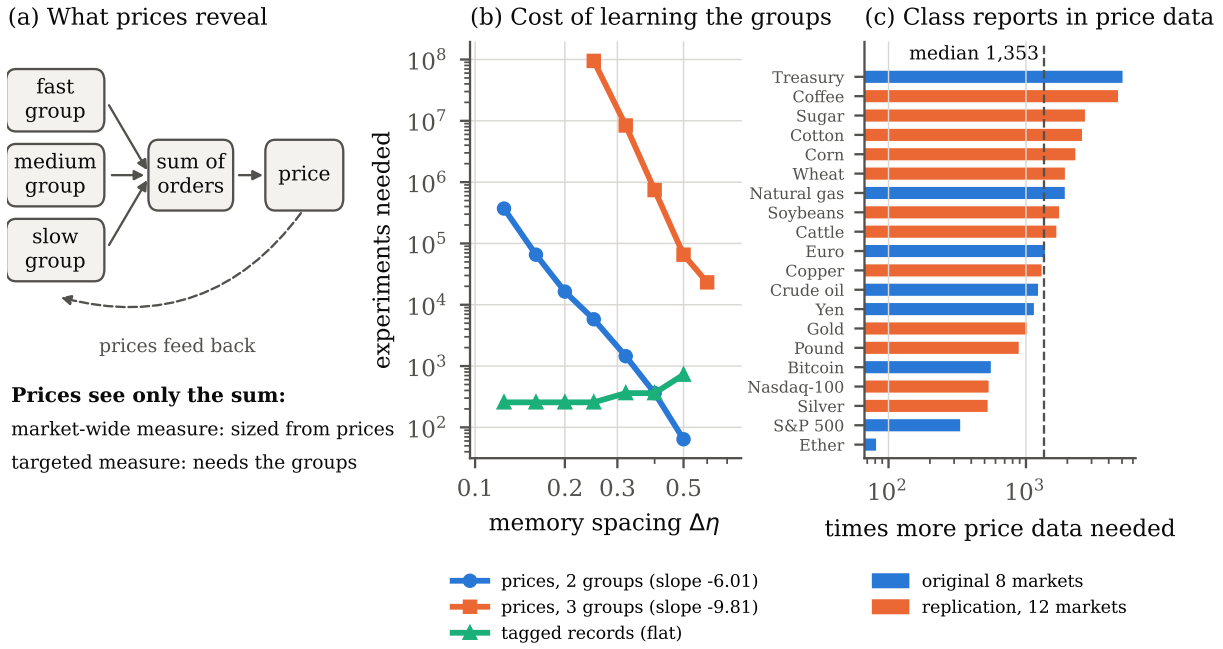


Figure 1: The paper in one figure. (a) Groups of traders reach the price only through the sum of their orders, so prices reveal the combined behaviour: enough to size a market-wide measure, not to decide whether halting one group is enough. (b) Pre-specified real-noise injection test (Supplement S16): experiments needed to recover the groups' weights against the spacing of their memories, with real one-minute Bitcoin price noise; price-only requirements grow with slopes −6.01 (two groups) and −9.81 (three groups; theory −6 and −10), tagged requirements stay flat. (c) Post hoc: how many times more data a price-only analysis would need to match the precision of the CFTC's weekly class report under standard root-T scaling, in the 20 markets of the registered tests (Table S26); the conversion assumes common root-T scaling and comparable estimands, which these data do not establish. Kind: registered results (b) and a calculation from registered results (c), `figs/make_fig_summary.py`.

the public-response equivalence class, an execution/capital interpretation under a specified probing contract, and a two-population supervisory decision whose feasible targeted action differs despite close public responses. The all-K action extension needs an additional weight-gap construction. Classical system identification (Ljung, 1999) provides the observation/estimation framework; a price-only record here is neither exogenous order flow nor a revealed portfolio preference. Novelty must be judged on these model-specific results against the cited theorems, not on AI terminology or a claim to have invented supervision without identification. Task-optimal exploration already supplies quantitative, task-dependent lower bounds and efficient experimental designs in linear systems (Wagenmaker et al., 2021). Accordingly, neither quantitative task-dependent learning nor choosing information for a decision is claimed as new here. Our comparison must turn on the latent population map, the

financial execution/capital contract, and the specified class intervention; no general dominance over task-optimal exploration is asserted.

Why prices hide the groups. The reason is the feedback loop. Algorithms trade on recent prices, their orders move prices, and those prices feed back into the algorithms. We model each group as tracking the price with a fading memory: recent prices count most, older prices count less, and one number, the group's memory rate, sets how fast the past fades. A desk behaves exactly this way if it uses a Kalman filter or a regularized online learning rule (Lemma 2.1). Because every group reaches the price only through the sum of all their trades, the price records the total and loses the parts. Think of several people pushing the same door: from the door's movement one can learn the total push and how it builds and fades, but two people who push in the same rhythm look like one stronger person.

Three ways to learn more. An observer that wants more than prices can wait and watch, which costs time and needs at least as many observations as an active prober needs experiments (Theorem 5.5); trade against the groups, which costs money, with capital rising as $1/\delta^4$ (Theorem S24.2); or acquire labelled records, which costs whatever the reporting system costs but removes the explosion in data (Theorem 6.2). Supplement S26 gives the details.

How large are the costs? In a hypothetical benchmark (Supplement S23), an outside trader that re-learned two unknown algorithms each week with test trades would lose at least \$15 million to \$356 million a year per market. Regulators do not trade to learn; the figure prices the information that labelled records deliver without any trading.

Why this matters for finance. Supervisors learn which data to require for which question, exchanges learn which questions their account-level data already answer, risk managers get a realistic limit on what can be inferred about other traders, and researchers get one reason why models with several trader types are hard to estimate (Supplement S20).

What the data show. Two real reporting systems let us check the main prediction, that class records pin down what prices leave open. The U.S. Commodity Futures Trading Commission collects positions from large futures traders and publishes a weekly snapshot of them, aggregated by trader category. In a pre-specified test in eight futures markets (CME Bitcoin and Ether, the E-mini S&P 500, 10-year Treasury notes, the euro, the yen, crude oil and natural gas), these class positions pin down the persistence of the designated speculative

class's positions to a range 9 to 71 times narrower than weekly prices allow, while prices alone leave it almost anywhere in the admissible range; a replication registered later on twelve new markets (metals, grains, softs, cattle, the Nasdaq-100 and the pound) holds in all twelve (Supplement S18). Converted into data, matching the precision of the weekly class report from prices alone would take, at the median market, about 1,350 times as much price data, the equivalent of thousands of years of weekly prices; this is hypothetical arithmetic, not a lower bound, because estimands and effective horizons differ and intervals reach parameter bounds. On Binance, in a pre-specified test on 30 markets never used before that matches the memory horizon of the two routes, the class data give the narrower interval in 29 of 29 evaluable markets, by a median factor of 7.3 (Supplement S19).

Passes and failures. Table S29 in the supplement lists all twenty-two empirical tests with their predictions and outcomes: eleven passed or held, ten failed and one secondary prediction was not supported. The primary CFTC result replicates on twelve new markets (12 of 12). The failures are part of the evidence: the CFTC's designated speculative class trend-followed in only four of the original eight markets (it did in eleven of the twelve new ones) and was the less persistent class in only three of the new twelve; the reports read as weekly flows did not pin down how persistent that flow is; positions showed no distinct faster root that could be learning memory; Binance prices pinned down the combined response in only 5 of 30 markets, and the combined response was far cheaper to learn than the separate rates in only 10 of 30 in a longer follow-up; class data did not improve return forecasts; and in October 2026 prices did not forecast the top-trader position beyond its own persistence (1 of 30), and taker flow did not pin down the combined response (5 of 30), although its uncertainty shrank at the predicted root-T rate in all 30. Supplement S21 reports the robustness checks.

Two supplementary results, on markets that share eigenvalues and price variance but not short-run risk and on how reduced-form feedback is measured, are summarized in Supplement S22.

Contribution. The candidate contribution is the market-model connection between a public-response equivalence class, the sample and financial resources needed for finer identification, and a specified supervisory action. Equal-memory groups are exactly inseparable; distinct but nearby groups face a finite-noise difficulty, not permanent nonidentification. Proposition 6.4 establishes a two-group separation example and a conditional general-K extension. The statistical clustering exponent and the distinction between task-relevant information and full identification have precedents; the precise comparison below states what

is inherited and what requires model-specific proof. Historical position-report evidence is descriptive and does not establish the action model or the necessity of granular reporting. Formal receipts cover particular typed statements; semantic correspondence and independent proof review remain separate obligations.

How to read the paper. Every main result is followed by a short paragraph headed "In words"; Supplement S28 gives a reading guide and the main symbols in plain words.

What we do not claim. The identification results hold for the stated linear model and for the stated observation contracts, that is, for what the observer is assumed to see. The model treats price impact and the persistence of pricing errors as known primitives rather than solving a dealer equilibrium, and it assumes the observer knows the fundamental value; real data meet neither assumption. They are not general limits on surveillance with participant-level records, which can be far more informative (Propositions S24.3 and 6.1). The matching upper bound holds on a class with separated rescaled rates and weights bounded away from zero. Capital figures depend on the observation-noise calibration, which we estimate rather than assume. The empirical results are reduced-form diagnostics; they do not identify structural learning parameters. Our pre-specified real-data test of the identification boundary failed its criterion on one of its three parts: on 42 days of data the merged response was precisely estimated in only 5 of 30 markets (Supplement S19). Binance's class ratios are positions, not tagged order flow, so they are the closest public analogue to the disclosure of Proposition 6.1, not an instance of it. We hold the algorithms' memories and intensities fixed across disclosure regimes; disclosure that reveals positions may lead informed desks to trade less or to split orders across accounts, which we do not model.

1.1 Relation to the literature

Three literatures meet here. In finance, learning algorithms and AI agents in markets are studied by simulating or modelling them (Colliard et al., 2022; Dou et al., 2025); traders who differ in horizon or speed are studied in equilibrium (Cespa and Vives, 2015; Biais et al., 2015; Crouzet et al., 2020); the exponentially fading beliefs we use are the extrapolative beliefs of Barberis et al. (2015, 2018); and heterogeneous-agent models estimated from prices often leave the type-separating parameters imprecise (Brock and Hommes, 1998; Amilon, 2008; Kukacka and Barunik, 2017), for which our results give one reason. Price formation follows the linear-impact tradition (Kyle, 1985; Glosten and Milgrom, 1985; Hasbrouck, 1991) and the transient-impact literature (Obizhaeva and Wang, 2013; Gatheral, 2010; Alfonsi et al., 2012); trader types have been inferred from public data, as in the probability of informed

trading (Easley et al., 1996), and from account data (Kirilenko et al., 2017). Learning by trading has an equilibrium counterpart in price experimentation (Leach and Madhavan, 1993), and execution that learns impact while trading is studied by Moallemi et al. (2008) and Park and Van Roy (2015). Technically, the closest work is closed-loop identification (Ljung, 1999; Forssell and Ljung, 1999), weak identification (Andrews and Cheng, 2012), and super-resolution and mixture estimation (Moitra, 2015; Heinrich and Kahn, 2018; Wu and Yang, 2020; Batenkov et al., 2021; Katz et al., 2023). We do not claim the K-group exponent as new: in mixture language it is the square of the condition number for K clustered nodes (Heinrich and Kahn, 2018; Wu and Yang, 2020), and Prony's method attains it (Katz et al., 2023); the δ^{-4} law for one against two close components is classical in resolution analysis (Shahram and Milanfar, 2005); logarithmic rates are known for mixtures of exponentials (Rebafka and Roueff, 2015); and labelled samples are known to be more informative than unlabelled ones (Castelli and Cover, 1996). The model-specific contribution proposed here is the closed-loop form (the memories are zeros, not poles, of the observed transfer function, so they must first be extracted from the loop), the passive equivalence, the translation of sample sizes into executed quantity, capital and money, and the value of class records, including a decision in which attribution changes the right action. Supplement S4 gives the full comparison, with tables of the closest results.

1.2 Outline

Section 2 sets out the model and Section 3 the identification boundary. Section 4 turns memory resolution into executed quantity, capital and money for a known pair, and Section 5 treats unknown mixtures, general memory and passive data. Section 6 states what disclosure buys and when it changes a supervisor's decision, and tests it on the CFTC's mandated trader-class reports and on Binance's class ratios. Section 7 draws the implications for supervision and Section 8 concludes. The online supplement contains the proofs, the extended literature review and the map of results, the discussion of the assumptions, general memory, the real-noise test, the trader-class tests in full, further results and robustness checks, the hypothetical dollar illustration, a finite-horizon decision example, the spectral blind spot, the measurement of reduced-form feedback, the formal-verification statement and the reproducibility map.

2 Model and motivating examples

The model in words. The market has one price and K groups of traders. Each period three things happen. First, the price moves: it drifts back toward its fundamental value, and it is pushed by the groups' trades, by any test trade and by news. Second, each group updates its view by mixing the newest price into its running average, with weight η_i on the newest price and $1-\eta_i$ on its old view. Third, each group trades in proportion to its view, with intensity q_i. A group with a large η_i has a short memory and reacts quickly, like a short-horizon trend follower; a group with a small η_i has a long memory and reacts slowly. The equations below write this down.

Time is discrete. Let p_t be the price, p^* the fundamental value and $x_t = p_t - p^*$ the pricing error. A reduced-form dealer sector absorbs net order flow with price impact $\kappa > 0$, an imposed primitive rather than an equilibrium quote, and arbitrage pulls the error toward zero with persistence $a \in [0, 1)$:

$$x_t = a x_{t-1} + \kappa \sum_{i=1}^{K} q_i y_{i,t-1} + \kappa v_t + w_t, \tag{1}$$

where v_t is an executed probe quantity and w_t a price innovation. Cohort i holds a state $y_{i,t}$ with exponential memory and trades $q_i y_{i,t}$:

$$y_{i,t} = (1-\eta_i)\, y_{i,t-1} + \eta_i x_t, \qquad \eta_i \in (0,1). \tag{2}$$

The ordering is *price first*: the price responds to the probe, then cohorts update. Write $r_i = 1-\eta_i$ for the memory rate and $w_i = q_i\eta_i$ for the merged weight. The feedback kernel and its transform are

$$k_m = \sum_i w_i r_i^m, \qquad G(z) = \sum_i \frac{w_i z}{z - r_i}, \tag{3}$$

and the transfer function from probes to prices, $H(z) = \sum_{j\geq 0} h_j z^{-j}$, is

$$H(z) = \frac{\kappa z}{z - a - \kappa G(z)} = \frac{\kappa z D(z)}{(z-a)D(z) - \kappa N(z)}, \qquad G = \frac{N}{D}, \quad D(z) = \prod_i (z - r_i), \tag{4}$$

with impulse response $h_0 = H(\infty) = \kappa, h_1, h_2, \ldots$; in particular $H(1) = \kappa/(1-S) = \sum_j h_j$. Stacking $z_t = (x_t, y_{1,t}, \ldots, y_{K,t})^\top$ gives $z_t = J z_{t-1} + b v_t$, where, because cohorts update on the same period's price, the probe enters every state on impact through

$$b = \kappa\,(1, \eta_1, \ldots, \eta_K)^\top, \tag{5}$$

and

$$J = \begin{pmatrix} a & \kappa q_1 & \cdots & \kappa q_K \\ \eta_1 a & 1-\eta_1+\eta_1\kappa q_1 & \cdots & \eta_1\kappa q_K \\ \vdots & & \ddots & \vdots \\ \eta_K a & \eta_K\kappa q_1 & \cdots & 1-\eta_K+\eta_K\kappa q_K \end{pmatrix}, \qquad S \equiv a + \kappa \sum_i q_i. \tag{6}$$

In the *positive economy* ($q_i \geq 0$), J is entrywise nonnegative, its price row sums to S, and the economy is Schur stable iff $S < 1$.

Lemma 2.1 (Two motivating examples: filtering and learning)**.** *(i)* Kalman memory. *A desk that tracks a latent random-walk trend $v_{t+1} = v_t + \zeta_{t+1}$, $\zeta \sim N(0, \sigma_\zeta^2)$, from signals $r_t = v_t + \nu_t$, $\nu \sim N(0, \sigma_\nu^2)$, using the steady-state Kalman filter, updates as $\hat{v}_{t+1} = (1-\eta)\hat{v}_t + \eta r_t$ with*

$$\eta = \frac{-\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2\sigma_\nu^2}}{2\sigma_\nu^2} \in (0,1). \tag{7}$$

(ii) Gradient memory. *Online gradient descent with L_2 regularization $\gamma \in (0,1)$, step size $\alpha > 0$ and loss $\frac{1}{2}H(x_t - \theta_t)^2$ gives $\theta_{t+1} = (1-\eta)\theta_t + \eta q x_t$ with $\eta = \gamma + \alpha H \in (0,1)$ and $q = \alpha H/\eta$. A tabular value update with learning rate η at a fixed state has the same form; general reinforcement-learning policies need not.*

Remark 2.2 (Why exponential memory). Lemma 2.1 shows that two different design philosophies, Bayesian filtering and regularized online learning, produce the same observable reduced-form belief state $y_{i,t}$. When desk i chooses order flow $u_{i,t}$ subject to $I_{i,t} = (1-\delta_{\text{inv}})I_{i,t-1} + u_{i,t}$ and an inventory cap $|I_{i,t}| \leq \bar{I}_i$ to maximize expected continuation return $\beta_i y_{i,t} u_{i,t}$ net of quadratic execution cost $\frac{1}{2}\lambda_{\text{tc},i}u_{i,t}^2$ and quadratic inventory holding cost $\frac{1}{2}\gamma_{\text{inv},i}I_{i,t}^2$, its interior first-order condition gives $u_{i,t}^* = q_i y_{i,t} - \psi_{\text{inv},i} I_{i,t-1}$ with market-response weight $q_i = \beta_i/(\lambda_{\text{tc},i}+\gamma_{\text{inv},i})$ and inventory mean-reversion $\psi_{\text{inv},i} = \gamma_{\text{inv},i}(1-\delta_{\text{inv}})/(\lambda_{\text{tc},i}+\gamma_{\text{inv},i})$, which shrinks toward zero as trading costs or inventory penalties tighten and destabilizes the market when the closed-loop spectral radius reaches one. This filtering-and-control recursion specifies an individual agent's decision rule, not a complete economic equilibrium (dealer–trader equilibrium is analysed in the companion paper).

Two further simplifications matter for reading the results: the observer is taken to know the fundamental value p^*, so the pricing error x_t is observed, and the dealer sector is a fixed linear impact κ rather than an equilibrium quote. The paragraph before Supplement S18 states what the data tests lose when neither holds. Assuming that the observer knows p^* gives it more information than any real observer has, so the lower bounds hold a fortiori

when p^* is unknown; Corollary 5.3 proves that baseline demeaning eliminates unknown p^* without affecting the upper-bound rate.

The model applies these rules to the pricing error x_t: a desk that cannot separate fundamental news from transitory deviations treats recent deviations as a signal of a persistent trend, which is the exponential extrapolation of Barberis et al. (2015, 2018). With $q_i > 0$ such a desk buys after rises, as trend followers do, and nothing in the model says this is profitable; the lemma shows only that the update has the exponential form. In these two illustrative families of algorithms the memory rate is a known function of a tuning parameter, the desk's signal-to-noise ratio or its regularization. That is the kind of private information a supervisor would like to recover, although real trading algorithms need not belong to either family. Section 3 shows why prices reveal it only in merged form.

Lemma 2.3 (Positive price-response geometry)**.** *In the positive, stable economy the impulse response is a nonnegative mixture of decaying exponentials,*

$$h_j = \sum_{\ell} \omega_\ell \lambda_\ell^j, \qquad \omega_\ell \geq 0, \quad \sum_{\ell} \omega_\ell = \kappa, \quad 0 \leq \lambda_\ell < 1.$$

It is completely monotone and nonincreasing; its peak is the direct impact κ, even though J is non-normal. In price/memory coordinates the price-only gain is $\sup_t \|e_x^\top J^t\|_2 = 1$.

As an illustration of the lemma (not part of its proof), $(-1)^n \Delta^n h_j \geq 0$ for $n \leq 4$ held in all 892 stable random positive economies tested. Lemma 2.3 matters twice: it makes probing costs positive (Section 4), and it shows that in this economy transient amplification of the price response cannot arise from cohort feedback alone. The companion paper locates amplification in order-book depth instead.

Assumptions. Supplement S14 discusses the six modelling choices behind the results: linearity, price-first timing within a period, exponential memory (relaxed in Supplement S17), algorithms that react to being probed, known primitives, and the observation contracts. Unknown primitives can only make separation harder, so the lower bounds remain valid; Corollary 5.3 proves that estimating (p^*, κ, a) at the parametric rate leaves the $\Theta((\Delta\eta)^{-(4K-2)})$ upper-bound rate unchanged.

2.1 The observer, its options and its costs

The results are stated from the side of one observer outside the trader groups: a supervisor, an exchange's surveillance unit or a large desk. Its *target* is attribution, recovering the merged weight and memory rate of each group to a stated accuracy, or deciding between two

candidate populations at a stated error. Before it observes, it knows the price persistence a, the impact κ and the law of the observation noise, and, for the K-group results, the number of groups; it does not know their rates or weights. These are the conditions under which the lower bounds are statements about the data rather than about the observer's ignorance; unknown primitives can only enlarge the set of candidate populations.

The observer has three options, and each result prices one of them.

(a) *Wait.* Record stationary prices driven by natural innovations, without trading. The cost is time: the number of observations needed (Theorems 5.4 and 5.5).

(b) *Probe.* Execute a sequence v_t of trades against the groups and record the price response. Probes execute at the average price of a linear book after the uncontrolled state update; the prober bears the execution loss $\frac{1}{2}v^\top T_h v$ plus fees (Lemma S24.1) and must fund margin throughout. The counted resource is the number of independent experiments, each starting from the reset state, or the executed quantity and capital of a single experiment (Theorems 4.1–S9.2).

(c) *Acquire disclosure.* Receive each class's flow or position, observed with noise, at an operating cost per supervisory cycle and a compliance cost per market (Proposition 6.1, Theorem 6.2, Proposition S27.1).

Within each period the order is: the probe (if any) executes and the price updates, the groups update their states on the new price, and the observer records the price with noise. The groups do not change their rules in response to being observed; Supplement S14 discusses what happens if they do. Probing experiments start from the reset state. A probe sent into a running market adds the free response of the current state, which by linearity superposes with the forced response; if the state is unknown it acts as a nuisance term, which can only make separation harder, so the lower bounds still hold, while the upper bounds as stated need the reset state or a known one. Independent resets are a benchmark experiment that no outside participant can run in a live market; we use them because they make the information content of the data exact, and the economic comparisons say whenever they rely on them. A numerical check without resets, in which a known random probe is sent into one long running market whose groups' state is unknown, keeps the $(\Delta\eta)^{-6}$ order for $K = 2$ and raises the constant by a factor of about 1.4 to 6 (Supplement S3.8). Regulators do not, in practice, trade to learn; the probing option describes an outside participant, or prices the information a supervisor would otherwise have to buy by trading.

Three money quantities appear and are kept apart: the *capital required*, the equity an account must hold so that its margin constraint is never breached while it probes (Theorem S24.2 and Lemma S24.4); the *round-trip trading loss*, the execution loss of a probe that

ends flat, which is what the account actually spends (Theorems S9.2, S9.3 and S9.4); and the *annualised loss* of renewing that probe weekly, used for the comparison with audit-trail budgets (Supplement S8). The lower bounds hold for the stated probe classes, deterministic probes and, in Theorem S9.3, non-anticipating sequential probes; they are not lower bounds on information obtained through other channels, which is the point of Section 6. Two rates recur and measure different things: δ^{-4} is the capital (and the average-power horizon) for a *known* symmetric pair whose rates differ by 2δ, while $(\Delta\eta)^{-(4K-2)}$, which is $(\Delta\eta)^{-6}$ for $K = 2$, is the number of experiments for K *unknown* groups with spacing $\Delta\eta$.

3 The identification boundary

A central question in algorithmic market surveillance is whether an observer of executions and prices can identify the individual strategies active in the market.

Theorem 3.1 (Merged rate–weight equivalence)**.** *Let $\kappa > 0$ and $a \in [0, 1)$ be known. Use zero initial states and a common probe contract, and observe the modeled price state x_t.*

(i) Equivalence class. *Two economies $\mathcal{E}^A = \{(q_i^A, \eta_i^A)\}$ and $\mathcal{E}^B = \{(q_j^B, \eta_j^B)\}$ produce identical price responses for every probe sequence if and only if their merged measures coincide:*

$$\sum_{i:\eta_i^A=\eta} q_i^A \eta \;=\; \sum_{j:\eta_j^B=\eta} q_j^B \eta \qquad \forall \eta \in (0, 1]. \tag{8}$$

Write $a' = a + \kappa M(\{0\})$, so that cohorts with $\eta = 1$ (rate 0) act as part of the anchor. For $a' \neq 0$ the minimal realization has dimension $1 + |\mathrm{supp}(M) \cap (0, 1)|$, counting distinct positive rates with nonzero merged weight after signed cancellation; for $a' = 0$ the price carries no state of its own and the dimension is $|\mathrm{supp}(M) \cap (0, 1)|$. A cohort with $\eta = 1$ merges into the anchor a without adding a pole.

(ii) Separation complexity under average power. *Consider testing a single cohort with rate $\bar{\eta}$ against two cohorts with rates $\bar{\eta} \pm \delta$ and the same total weight, on a compact family that is uniformly Schur stable with all rates in $[\epsilon, 1 - \epsilon]$ for some $\epsilon > 0$. Observe $\tilde{x}_t = x_t + \nu_t$ with $\nu_t \sim N(0, \sigma_\nu^2)$ and use deterministic probes with $\frac{1}{T}\sum_{t<T} v_t^2 \leq P$. Any test with worst-case error at most $\alpha \in (0, 1/4)$ needs*

$$T \;\geq\; \frac{4\sigma_\nu^2(1 - 2\alpha)^2}{C_{\mathcal{F}}^2 P\, \delta^4} = \Omega(\delta^{-4}), \tag{9}$$

where $C_{\mathcal{F}}$ depends only on the family, through ϵ, the bound on $|H|$ implied by uniform stability, and the total weight q.

In words. If two trading desks update on prices with the same memory, no amount of price data can tell them apart or say how much each trades: prices reveal only how much trading happens at each memory speed. Desks with different memories can be told apart in principle, and the rest of the paper prices how hard that is.

The memory rates are poles of G and *zeros* of H, not closed-loop poles. This has a practical consequence we verify in Section 5: pole-fitting methods applied directly to the price impulse response recover closed-loop eigenvalues (0.4256 and 0.8224 for $K = 1$, $\eta = 0.3$, $a = 0.5$, $\kappa = 0.8$, $q = 0.2$) and miss the memory rate 0.70.

Remark 3.2 (Scope). The result concerns the stated model and observation contract. Cohorts sharing an exact rate enter prices through their merged weight; this does not determine identities or wealth allocations. Nearby rates can be distinguishable with enough information. The δ^{-4} statement in (ii) is a lower bound on a declared family, with family-dependent constants. Individual algorithms may choose their rates strategically, but any unilateral change maps into the same observer-facing merged measure, which reinforces rather than circumvents the boundary.

Proposition 3.3 (Conditional endogenous-pricing invariance)**.** *Let $\kappa = \mathcal{K}(\Gamma_M(\kappa))$ on a common admissible interval, with identical innovation laws, initial-state conventions and other primitives across economies. Here $\Gamma_M(\kappa)$ is the specified covariance record at a fixed impact and $\mathcal{K}$ is a maintained dealer rule, not a derived general equilibrium.*

(i) If equal merged measures imply the same map Γ_M and the fixed point is unique (or a common selection rule depends only on that map), equal merged measures give equal selected impacts.

(ii) Suppose $\mathcal{K}$ is L_K-Lipschitz, each covariance map is L_Γ-Lipschitz in impact, $L_K L_\Gamma < 1$, and $\sup_\kappa \|\Gamma_A(\kappa) - \Gamma_B(\kappa)\| \le \epsilon$. For existing fixed points,

$$|\kappa_A - \kappa_B| \le \frac{L_K \epsilon}{1 - L_K L_\Gamma}. \tag{10}$$

Thus a covariance discrepancy of order $(\Delta\eta)^{2K-1}$ preserves that order only under these uniform conditions.

Limits of the pricing extension. Anonymous flow alone implies neither a covariance-only dealer objective nor uniqueness or continuity of its equilibrium. Multiple equilibria, threshold quote changes, private information and inventory-dependent rules are outside this proposition. Preserving a price-record testing bound additionally requires a uniformly Lipschitz observation law with the same nondegenerate noise; adding anonymous flow is a

different statistical experiment and requires its own divergence bound. We therefore make no unconditional claim that endogenous pricing or additional aggregate-flow observations preserve the price-only lower bound.

4 Probing resources, memory resolution and money

Observation length alone cannot buy identification when the lifetime quantity of controlled executions is fixed. This section connects the merged-memory experiment to a financially specified input/output contract. Throughout, Φ is the standard normal distribution function.

4.1 Quantity

Theorem 4.1 (Sharp probing-resource orders for a known symmetric pair)**.** *Fix $0 \le a < 1$, $\kappa, q > 0$, $S = a + \kappa q < 1$ and $\eta \in (0,1)$. Compare a single cohort of weight q and rate η with two cohorts of weight $q/2$ and rates $\eta \pm \delta$, both in $[\epsilon, 1-\epsilon]$. All parameters are known. The price receives executed quantity through κv_t, starts from zero, has no process noise, and is observed with independent $N(0,\sigma^2)$ noise. Probes are deterministic and common to both alternatives. Let $d_T = (h^\delta_j - h^0_j)_{j<T}$ and $K_* = 2\kappa^2 q/[\epsilon^3(1-S)^2]$.*

(i) For every probe with $\sum_t |v_t| \le V$,

$$D_{\mathrm{KL}}(P^v_0 \| P^v_\delta) \le \frac{K_*^2 V^2 \delta^4}{2\sigma^2}, \qquad e_{\mathrm{worst}} \ge \frac{1}{2}\Big(1 - \frac{K_* V \delta^2}{2\sigma}\Big)_+. \tag{11}$$

(ii) The optimal worst-case error over these probes and all classifiers is exactly

$$e^*_T(V) = \Phi\left(-\frac{V\|d_T\|_2}{2\sigma}\right), \tag{12}$$

attained by a pulse at the first input and the likelihood-ratio test. Waiting adds information but $\|d_T\|_2 \uparrow \|d_\infty\|_2 < \infty$, so it cannot make the error arbitrarily small at fixed V. Error at most $\alpha < 1/2$ requires $V \ge 2\sigma(1-2\alpha)/(K_*\delta^2)$ independently of T.

(iii) The first two taps coincide and $d_2 = -\kappa^2 q \delta^2$. A single pulse of size

$$A = \frac{2\sigma\Phi^{-1}(1-\alpha)}{\kappa^2 q \delta^2} \tag{13}$$

and the third observation achieve error exactly α. The necessary and sufficient quantity order is δ^{-2}.

(iv) Under average power $\sum_{t<T} v_t^2 \le PT$*, the necessary horizon order is* δ^{-4}*, attained by the pulse whenever* $T \ge \max\{3, \lceil A^2/P \rceil\}$*.*

(v) For a fixed monetary price exposure $\lambda \neq 0$*, additive reserve* L *and cost envelope* $c\sum|v_t|$*, the sharp robust quantity allowance under budget* $B > L$ *is* $V_B = (B - L)/(|\lambda|\kappa + c)$*.*

In words. To tell one desk from two desks whose memories differ slightly, the observer must trade against them, and the quantity it must trade grows like $1/\delta^2$: halving the difference quadruples the trading needed. Waiting longer after a fixed trade does not help, because the trade's echo in the price dies out.

The identity in (iii) follows by direct substitution, and code checks it at $\delta \in \{0.02, 0.05, 0.10\}$: the difference is -1.2×10^{-4}, -7.5×10^{-4} and -3.0×10^{-3} at $\kappa = 1$, $q = 0.3$, and -3.2×10^{-4} at the calibration $\kappa = 0.8$, $q = 0.2$, $\delta = 0.05$. The theorem does not solve estimation of an unknown mixture (Section 5).

A Monte Carlo check of the exact error. Formula (12) is exact for the likelihood-ratio test. We simulated 200,000 noisy paths per hypothesis at the calibration ($\sigma = 0.01$, $\delta = 0.05$) for several pulse sizes and horizons (Table S32); simulated errors match the formula within Monte Carlo standard error under both hypotheses.

Figure S7 shows why the problem is hard. The two economies have the same direct impact and the same first-lag response; they first differ at lag two, by an amount proportional to δ^2, and the whole difference sequence scales with δ^2. A test therefore needs the signal-to-noise ratio $V\|d\|/\sigma$ to grow like δ^{-2}, which is the quantity order, and because the adverse mark of a probe grows with the square of its size, capital grows like δ^{-4}.

4.2 Execution cost and capital

Supplement S24 puts the quantity bound into money. Executing a probe against a linear book costs a positive-definite quadratic form in the probe, so no round trip yields a sure gain (Lemma S24.1). A counterparty that fills the probe needs margin that grows with the square of the probe, so the capital needed grows as $1/\delta^4$: halving the difference between two memories multiplies it by sixteen (Theorem S24.2); the same order holds for capital paid by the probe trader (Lemma S24.4). If the observer can instead see an account's exact equity, one test trade settles the question (Proposition S24.3): the difficulty lies in what the observer is allowed to see, not in the market.

A worked example with two Kalman desks that differ only in their signal-to-noise ratios (Supplement S15) puts these orders into numbers.

5 Unknown mixtures and passive data

Theorem 4.1 concerns a *known* pair. A supervisor faces an unknown number of cohorts with unknown rates and weights. This section asks how fast the cost of recovering them grows.

Two facts from the case of two cohorts carry over (Supplement S25). Fitting poles to the price response targets the closed-loop eigenvalues and misses the memory rates, which are zeros of the transfer function. And for two unknown groups no estimator, biased or not, separates them with fewer than order $(\Delta\eta)^{-6}$ experiments (Theorem S25.1), while the two hardest-to-separate economies have the same summed price response, so what prices cannot see carries no market-wide risk (Remark S25.2). The general case follows.

5.1 Many unknown cohorts

Theorem 5.1 (K-cohort two-point lower bound)**.** *Fix $K \geq 2$, a center $r_c \in (0,1)$, total merged weight $W > 0$ and $0 < \Delta\eta < \min(r_c, 1 - r_c)$, so that all rates lie in $(0,1)$, and assume $a + \kappa W/(1 - r_c - \Delta\eta) < 1$, so that every candidate economy is Schur stable. Each reset experiment observes $T \geq 2K$ noisy taps ($T = \infty$ allowed); the bound below uses the full response norm, which dominates any finite prefix. There exist two K-cohort economies with rates in $r_c + \Delta\eta[-1, 1]$, positive weights, and merged measures whose moments of order $0, \ldots, 2K-2$ coincide while the moment of order $2K-1$ differs by a constant times $(\Delta\eta)^{2K-1}$. Their impulse responses differ by $\|h^A - h^B\|_2 \asymp (\Delta\eta)^{2K-1}$, so recovering the weights from N independent noisy reset experiments with worst-case error probability below $1/4$ requires*

$$N \;\geq\; c\,\sigma^2 (\Delta\eta)^{-(4K-2)}.$$

The same requirement holds for the permutation-invariant loss $W_1(\hat{M}, M)$ between estimated and true merged measures at accuracy $c_K W \Delta\eta$, where $2c_K > 0$ is the Wasserstein distance between the two normalized quadrature rules, so no relabelling of atoms escapes it.

The two losses have different targets. The weights of the two economies differ by a constant that does not shrink with $\Delta\eta$, so the first statement is about weights at fixed accuracy, which Theorem 5.2 matches; the Wasserstein statement is about the memory distribution at accuracy proportional to $\Delta\eta$, which Theorem S10.1 matches.

The construction uses the K-point Gauss and Gauss–Radau rules of the same weight function: both are positive, have K atoms and integrate polynomials of degree $2K - 2$ exactly. For $K = 2$ it reproduces the order of Theorem S25.1. Table S33 lists the two rules: their low moments agree to machine precision and their moment of order $2K - 1$ differs

by 0.222, 0.0533 and 0.0131 for $K = 2, 3, 4$. The shrinking gap explains why constants deteriorate with K on top of the exponent.

What counts as an experiment. In Theorem 5.1 an experiment is a unit impulse from the reset state whose response is observed with noise of variance σ^2 per tap. The exponent is the same for any probe design of bounded energy. For any probe v the mean difference between the two economies is at most $\sup_\omega |\Delta\hat{H}(e^{i\omega})|\,\|v\|_2$, and $\sup_\omega |\Delta\hat{H}| = O((\Delta\eta)^{2K-1})$ because $\Delta\hat{H} = z^{-1}\hat{H}_A\hat{H}_B\,\Delta G$ with both responses bounded on the circle. Experiments with probe energy E each therefore need $NE \gtrsim \sigma^2(\Delta\eta)^{-(4K-2)}$: a larger probe buys the same information with fewer experiments, and Supplement S9 prices the capital that a larger probe requires. The bound does not cover information obtained without trading against the groups, such as class records (Section 6).

Table S34 and Figure S8 confirm the order numerically.

Theorem 5.2 (Matching upper bound for every K)**.** *Fix $K \geq 2$, a compact set $\mathcal{S} \subset \{s \in [-1,1]^K : s_1 < \cdots < s_K\}$ of ordered rescaled rates with $\min_{i\neq j}|s_i - s_j| \geq \gamma > 0$, and merged weights $w_i \in [\omega_{\min}, \omega_{\max}]$ with $\omega_{\min} > 0$. Consider the class of K-cohort economies with rates $r_c + \Delta\eta\, s$, $s \in \mathcal{S}$, and N independent reset experiments whose responses carry Gaussian noise with variance σ^2. There are constants $c_0, C > 0$ depending only on $(K, \mathcal{S}, \gamma, \omega_{\min}, \omega_{\max}, a, \kappa, r_c)$ such that, whenever $\epsilon_N \equiv \sigma N^{-1/2}(\Delta\eta)^{-(2K-1)} \leq c_0/\sqrt{\log(4K/\beta)}$, Prony's method applied to the first $2K$ estimated kernel values, with its output atoms listed in increasing order of rate, recovers the weights with*

$$\max_i |\hat{w}_i - w_i| \;\leq\; C\sqrt{\log(4K/\beta)}\;\epsilon_N$$

with probability at least $1-\beta$. Hence $N = C'\sigma^2(\Delta\eta)^{-(4K-2)}$ experiments suffice for any fixed accuracy and confidence. On any such class that contains the Gauss and Gauss–Radau pair of Theorem 5.1 (a rich class), together with that theorem the minimax sample requirement is $N^* \asymp \sigma^2(\Delta\eta)^{-(4K-2)}$. On smaller classes the upper bound need not be tight: if the rates are known, for example, the weights need only order $\sigma^2(\Delta\eta)^{-2}$ experiments. The constants c_0 and C grow quickly with K, through the conditioning of the moment map; the theorem gives the order in $\Delta\eta$ for fixed K, not a usable constant for large K (in Supplement Table S4 our double-precision Prony implementation already fails at $K = 4$ on the disclosure grid).

Proof. The kernel values $k_m = \sum_i w_i r_i^m$ are the power moments of the merged measure. The closed-loop recursion $h_t = ah_{t-1} + \kappa\sum_m k_m h_{t-1-m}$ (plus κ at $t = 0$) can be solved for $k_0, \ldots, k_{2K-1}$ from $h_1, \ldots, h_{2K}$. Because $h_0 = \kappa$ is a known primitive, each k_m is a polynomial in $h_1, \ldots, h_{m+1}$ with coefficients depending on (a, κ) (the only division is by $\kappa h_0 = \kappa^2$), so

on the bounded set of responses within sup-distance 1 of the class this map is L_h-Lipschitz, with L_h depending on the class but not on $\Delta\eta$. Let E be the event $\max_{1\le t\le 2K}|\bar{h}_t - h_t| \le \sigma N^{-1/2}\sqrt{2\log(4K/\beta)}$, where $\bar{h}_t$ averages the N responses; a Gaussian tail bound and a union bound over the $2K$ estimated taps give $P(E) \ge 1-\beta$. On E, and provided this radius is at most 1, the estimated kernel values satisfy $\max_m |\hat{k}_m - k_m| \le L_h\sigma N^{-1/2}\sqrt{2\log(4K/\beta)}$. Centring at r_c is a fixed triangular linear map, $\mu_m = \sum_{j\le m}\binom{m}{j}(-r_c)^{m-j}k_j$, and $\mu_m = (\Delta\eta)^m t_m$ with $t_m = \sum_i w_i s_i^m$. The map $F : (s,\omega) \mapsto (t_0,\dots,t_{2K-1})$ does not depend on $\Delta\eta$; its Jacobian is a confluent Vandermonde matrix with determinant $\prod_i \omega_i \prod_{i<j}(s_i - s_j)^4$, bounded away from zero on the compact class. On ordered, γ-separated nodes F is injective, because a measure with K distinct atoms is determined by its first $2K$ moments. A C^1 map that is injective on a compact set Q and has a nonsingular Jacobian there is injective on an open neighbourhood U of Q: otherwise there are $x_n \ne y_n$ with $F(x_n) = F(y_n)$ and distance to Q tending to zero, whose limits $x, y \in Q$ satisfy $F(x) = F(y)$, so $x = y$, contradicting local injectivity at x. Applying the quantitative inverse function theorem uniformly on Q, there are $\rho_0 > 0$ and $L < \infty$, uniform over the class, such that $F(U)$ contains the ball of radius ρ_0 around every point of $F(Q)$ and the single inverse $F^{-1} : F(U) \to U$ is L-Lipschitz on each such ball. An error e in μ_m is an error $e(\Delta\eta)^{-m}$ in t_m, largest at $m = 2K-1$. Since $\Delta\eta < 1$ and the centring map is fixed, on E we get $\|\hat{t} - t\|_\infty \le C_1\sqrt{\log(4K/\beta)}\,\epsilon_N$ with C_1 equal to $\sqrt{2}L_h$ times the norm of the centring map; and because $\sigma N^{-1/2}\sqrt{2\log(4K/\beta)} \le \sqrt{2}\,\epsilon_N\sqrt{\log(4K/\beta)}$, the radius condition holds whenever $c_0 \le 1/\sqrt{2}$. When this is below ρ_0, $\hat{t}$ lies in the ball where F^{-1} is L-Lipschitz. For an input in such a ball, $x = F^{-1}(\hat{t})$ has K distinct nodes and nonzero weights, so the $K \times K$ Hankel matrix of $\hat{t}$ is nonsingular and Prony's method returns exactly x: the Prony polynomial is determined uniquely by the Hankel system, and since the K real nodes of x reproduce $\hat{t}$ it equals $\prod_i(z - \hat{s}_i)$, so its roots are real and distinct and no complex pair can appear; Prony's method computes the algebraic inverse of the moment map up to the order of the atoms, so, listing its output in increasing order of rate (which the error bound below keeps unambiguous once $C\sqrt{\log(4K/\beta)}\,\epsilon_N < \gamma/2$), its output is $F^{-1}(\hat{t})$ and the stated bound follows with $C = LC_1$ and $c_0 = \min\{\min(\rho_0, \gamma/(2L))/C_1,\, 1/\sqrt{2}\}$, which also keeps $C\sqrt{\log(4K/\beta)}\,\epsilon_N < \gamma/2$. No separate root-perturbation bound is needed: on the ball where F^{-1} is L-Lipschitz, Prony's roots and weights are the coordinates of $F^{-1}(\hat{t})$, so the conditioning of the root step is part of L (for an explicit step-by-step analysis of Prony's method with clustered exponents see Katz et al., 2023). The matching statement requires the class to contain the Gauss and Gauss–Radau pair of Theorem 5.1 (ordered nodes with separation at least the smaller of their two minimum gaps, and weights within $[\omega_{\min}, \omega_{\max}]$ after scaling to total weight W); for any such class the orders match, with constants that depend on the class. □

In words. With K unknown groups whose memories lie within $\Delta\eta$ of each other, the number of experiments needed grows like $(\Delta\eta)^{-(4K-2)}$. For two groups, halving the spread multiplies the data needed by 64; for three groups, by 1,024. Theorem 5.1 shows that no method does better, and this theorem shows that Prony's method achieves it.

Machine check. Lean verifies this theorem for every $K \geq 1$, using only Lean's standard axioms. The formal proof follows the one above: a Lipschitz bound for the closed-loop deconvolution that does not depend on $\Delta\eta$, the $(\Delta\eta)^{-m}$ rescaling, a sub-Gaussian tail bound over the $2K$ taps, injectivity of the Jacobian of the moment map (proved by Hermite interpolation, without the determinant formula), and a uniform inverse function theorem on the compact class. The formal estimator is the representation with increasing nodes and nonzero weights that matches the $2K$ estimated moments; Lean shows that it is unique and that Prony's linear system has a unique solution whose polynomial has exactly its nodes as roots, so the roots are real. The formal statement allows sub-Gaussian noise and assumes $|r_c| + \Delta\eta \leq 1$, so that every memory rate has modulus at most one.

Corollary 5.3 (Unknown primitives cost nothing in the rate)**.** *In the setting of Theorem 5.2, suppose in addition that every memory rate is at least $\varepsilon > 0$ (that is, $r_c - \Delta\eta \geq \varepsilon$), and that the fundamental value p^*, the impact $\kappa \in [\underline{\kappa}, \overline{\kappa}] \subset (0, \infty)$ and the persistence $a \in [0, \bar{a}] \subset [0, 1)$ are unknown. Put $\tilde{k}_0 = k_0 + a/\kappa$ and $\tilde{k}_m = k_m$ for $m \geq 1$.*

(i) The response depends on (a, κ, M) only through κ and $\tilde{k}$: $h_0 = \kappa$ and $h_{t+1} = \kappa \sum_{m=0}^{t} \tilde{k}_m h_{t-m}$. The persistence a acts exactly like an extra group with memory rate zero and merged weight a/κ.

(ii) The taps $h_0, \ldots, h_{2K+1}$ identify (κ, a, M): $\kappa = h_0$; the recursion gives $\tilde{k}_0, \ldots, \tilde{k}_{2K}$; the values $\tilde{k}_1, \ldots, \tilde{k}_{2K}$ are the moments of orders $0, \ldots, 2K-1$ of $\mu' = \sum_i w_i r_i \delta_{r_i}$; and $a = \kappa(\tilde{k}_0 - \sum_i w_i)$.

(iii) Subtract the pre-probe price to remove p^* (doubling the noise variance), estimate $\hat{\kappa} = \hat{h}_0$, deconvolve with $\hat{\kappa}$ alone, apply Prony's method to the estimated moments of μ', and set $\hat{w}_i = \widehat{w_i r_i}/\hat{r}_i$ and $\hat{a} = \hat{\kappa}(\hat{\tilde{k}}_0 - \sum_i \hat{w}_i)$. There are constants $c_0, C > 0$ depending on $(K, \mathcal{S}, \gamma, \omega_{\min}, \omega_{\max}, \varepsilon, \underline{\kappa}, \overline{\kappa}, \bar{a}, r_c)$ but not on $\Delta\eta$ such that, whenever $\epsilon_N \leq c_0$,

$$\max_i |\hat{w}_i - w_i| + |\hat{a} - a| + |\hat{\kappa} - \kappa| \;\leq\; C\sqrt{\log(8K/\beta)}\,\epsilon_N$$

with probability at least $1 - \beta$. Hence $N = C'\sigma^2(\Delta\eta)^{-(4K-2)}$ experiments suffice, the same order as with known primitives.

In words. Not knowing the market's own persistence or its price impact costs a constant factor, not a worse rate. Persistence enters the price exactly like one more group of traders with no memory at all, and that group sits far from the clustered groups the supervisor is trying to separate. Measured in the merged weights times the rates, the clustered groups' problem is the same Prony problem as before, with weights bounded below because rates are bounded away from zero; one more tap pins down the extra group.

Proof of Corollary 5.3. (i) Substitute $k_0 = \tilde{k}_0 - a/\kappa$ in $h_{t+1} = ah_t + \kappa \sum_{m \le t} k_m h_{t-m}$: the term ah_t cancels against $\kappa(a/\kappa)h_t$. (ii) Solving the recursion of (i) for $\tilde{k}_t$ gives $\tilde{k}_t = (h_{t+1} - \kappa \sum_{m<t} \tilde{k}_m h_{t-m})/\kappa^2$, a polynomial in $h_1, \dots, h_{t+1}$ divided by a power of $h_0 = \kappa$. For $m \ge 1$, $\tilde{k}_m = \sum_i w_i r_i^m = \sum_i (w_i r_i) r_i^{m-1}$, so $\tilde{k}_1, \dots, \tilde{k}_{2K}$ are the moments of orders $0, \dots, 2K-1$ of μ', which has K distinct atoms with positive weights; these moments determine μ' (Theorem 3.1). Since $r_i \ge \varepsilon > 0$, $w_i = (w_i r_i)/r_i$, and then $a = \kappa(\tilde{k}_0 - \sum_i w_i)$. (iii) On the event that every estimated tap is within $\sigma N^{-1/2}\sqrt{2\log(8K/\beta)}$ of its mean (probability at least $1 - \beta/2$ by the Gaussian tail and a union bound over $2K+2$ taps, with the doubled variance absorbed into the constant), $\hat{\kappa} \ge \underline{\kappa}/2$ once $\epsilon_N \le c_0$, so the map from taps to $\hat{\tilde{k}}$ is Lipschitz with a constant that depends on $(K, \underline{\kappa}, \overline{\kappa}, \bar{a}, W)$ and not on $\Delta\eta$. The weights of μ' lie in $[\varepsilon\omega_{\min}, \omega_{\max}]$ and its rates are those of the class of Theorem 5.2; the proof of that theorem uses only a $\Delta\eta$-free Lipschitz deconvolution, the rescaling by $(\Delta\eta)^{-m}$ and weights bounded below, so it applies verbatim to μ' and gives $\max_i |\widehat{w_i r_i} - w_i r_i| \le C_1\sqrt{\log(8K/\beta)}\,\epsilon_N$ and rate errors smaller by a factor $\Delta\eta$, with probability at least $1 - \beta/2$. The maps $(v, r) \mapsto v/r$ on $r \ge \varepsilon/2$ and $(\kappa, \tilde{k}_0, w) \mapsto \kappa(\tilde{k}_0 - \sum_i w_i)$ are Lipschitz on the bounded set reached, and $|\hat{\kappa} - \kappa|$ and $|\hat{\tilde{k}}_0 - \tilde{k}_0|$ are of order $\sigma N^{-1/2} \le \epsilon_N$. Combining gives the bound. □

Table S35 confirms the order numerically. For numerical stability the estimator works in centred and rescaled coordinates (the estimator is the same; only its floating-point evaluation changes). The argument parallels the stability analysis of super-resolution for clustered complex nodes (Batenkov et al., 2021), and Katz et al. (2023) show that Prony's method attains the min-max bound for clustered exponents; here the nodes are real and are first extracted from the closed loop, and it reproduces the ESPRIT slope of -6.15 reported for $K = 2$ in the earlier version.

Real market noise. Theorems 5.1 and 5.2 assume Gaussian noise. A pre-specified test (Supplement S16) keeps the trader groups synthetic and replaces the noise with heavy-tailed one-minute Bitcoin price changes: the price-only requirement still grows with fitted slopes -6.01 for $K = 2$ and -9.81 for $K = 3$ (theory -6 and -10), while the tagged requirement stays bounded across the spacing grid.

General memory. If memory is any completely monotone kernel, a finite or infinite mixture of exponential memories, recovery becomes far harder: the minimax accuracy for the memory distribution is of exact order $1/\log(N/\sigma^2)$, so reaching accuracy w requires and suffices $N/\sigma^2 = \exp(\Theta(1/w))$ (Theorem S17.2, Supplement S17). The polynomial rates above rely on memory being a finite mixture of exponentials.

5.2 Passive data

Theorem 5.4 (Passive identification: minimum phase)**.** *Consider a stable economy with $0 < a < 1$, distinct memory rates $0 < r_i < 1$ and nonzero merged weights.*

(i) The zeros of H are the memory rates r_i together with a structural zero at $z = 0$; none cancels against the denominator, and all lie in the open unit disk, so H is minimum phase. The degenerate anchor ($a = 0$) and zero-rate cases require the cancellations described in the proof of Theorem 3.1.

(ii) Let the price be driven by a single innovation sequence with variance $\sigma_\xi^2 > 0$, possibly unknown, so that its spectrum is $f(\omega) = \sigma_\xi^2 |H(e^{i\omega})|^2/(2\pi)$. Within the class of (i), f determines $\sigma_\xi^2\kappa^2$ and the normalized response H/h_0 (recall $h_0 = \kappa$); hence it determines the memory rates. If in addition κ (or σ_ξ^2) and a are known, it determines H and, by Theorem 3.1, the merged measure.

Machine check. For every K, Lean checks part (i) and part (ii) starting from equal spectra at every frequency (`p1_thm_minphase_zeros`, `p1_thm_minphase_spectrum_H`, `p1_thm_minphase_spectrum_merged`). The check uses H in the rational form $\kappa z D(z)/((z - a)D(z) - \kappa N(z))$ of Theorem 3.1, and *stable* in the sense that the poles of H lie in the open unit disk. Both are derived from the market recursion. The eigenvalues of the transition of the state $(x_t, y_{1,t}, \ldots, y_{K,t})$ are exactly the zeros of the denominator of H, so Schur stability of the recursion is the pole condition (`p1_eigen_iff_bottom_root`, `p1_schur_stable_iff_poles_in_disk`). The generating function of the recursion's price response h_t to a unit trade satisfies $\left(\sum_t h_t x^t\right) x^{K+1} B(1/x) = x^{K+1}\kappa x^{-1} D(1/x)$ as formal power series, where B is the denominator of H; with $x = 1/z$ this is $\sum_t h_t z^{-t} = H(z)$ (`p1_impulse_generating`, `p1_tfBottomRev_eval`, `p1_tfTopRev_eval`).

The result holds for signed weights (mean-reverting cohorts): in 1,733 stable random economies with weights of both signs, the nonstructural zeros equalled the memory rates and lay inside the unit disk. With a second independent shock in the cohorts' memories, the Jacobian of the price spectrum with respect to six model parameters has full rank, with

singular values from 34.51 down to 0.0698. Passive noise helps identification rather than hindering it.

Theorem 5.5 (Passive cost)**.** *Suppose the price is observed passively as the stationary output $x_t = H_\theta(L)\xi_t$ of i.i.d. Gaussian innovations $\xi_t \sim N(0, \sigma^2)$, $\sigma^2 > 0$ known, without observation noise, for θ in a compact, uniformly Schur-stable family with rates in $[\eta_{\min}, \eta_{\max}] \subset (0, 1)$, and let $D(\theta_1 \| \theta_0)$ be the Kullback–Leibler divergence rate per observation.*

(i) For a single cohort at $\bar{\eta}$ against the symmetric pair at $\bar{\eta} \pm \delta$ with the same total weight, $D = \Theta(\delta^4)$.

(ii) For the moment-matched two-cohort economies of Theorem S25.1, $D = \Theta(\tau^2(\Delta\eta)^6)$.

(iii) For every $K \geq 2$, for the K-cohort Gauss and Gauss–Radau economies of Theorem 5.1, $D = \Theta((\Delta\eta)^{4K-2})$.

Hence any test based on T passive observations with worst-case error at most $\alpha < 1/4$ needs $T = \Omega(\delta^{-4})$, $T = \Omega(\tau^{-2}(\Delta\eta)^{-6})$ and $T = \Omega((\Delta\eta)^{-(4K-2)})$ respectively, uniformly as the separation shrinks: passive data cost at least the same polynomial order as active probing for every number of cohorts. For each fixed pair, at fixed type-I error the likelihood-ratio test's type-II error decays at exponential rate D (Stein's lemma), and both errors decay at the Chernoff-information rate, which is of the same order in the separation; so the orders are the right ones; we do not claim that this attainment is uniform as the separation shrinks. All constants depend only on K, the stability margin of the family, the rate interval $[\eta_{\min}, \eta_{\max}]$, the total weight and the moment-matched construction; they do not depend on δ, $\Delta\eta$ or T. In part (ii) τ is held fixed as $\Delta\eta \to 0$. The testing conclusions use a finite-record bound $D_{\mathrm{KL}}(P_1^T \| P_0^T) \leq C\,T\,\varepsilon^2$, with ε the leading moment gap (δ^2, $\tau(\Delta\eta)^3$, $(\Delta\eta)^{2K-1}$); the lower bounds on D and the fixed-pair error exponents are separate statements.

Machine check. Lean checks the finite-sample consequence of parts (i) and (iii), not the divergence-rate statements. In the checked record the market starts from rest and runs for any number M of warm-up periods before the T recorded prices. The bound does not depend on M, but the passage to the stationary law ($M \to \infty$) is not checked. Part (iii) is checked for every pair of populations with nonnegative merged weights of total at most W, normalized rates in $[-1, 1]$ and equal moments below order $2K - 1$ (`p1_thm_passive_cost_matched`). This covers the K-cohort Gauss and Gauss–Radau economies of Theorem 5.1: both rules are positive with nodes in $[-1, 1]$ and integrate polynomials of degree $2K - 2$ exactly. Lean proves these properties of the two rules for $K = 2$ and $K = 3$ (`p1_thm_passive_cost_gauss_radau_two`,

`p1_thm_passive_cost_gauss_radau_three`), so for these K the check covers the paper's economies with no extra hypothesis. For general K, the exactness of the two quadrature rules, a classical property, enters the check as the moment hypothesis. It also covers the binomial pair used in the numerical checks (`p1_thm_passive_cost_every_K`).

In words. Watching the market without trading does not escape the barrier. A passive observer needs at least the same polynomial order of observations as a prober needs experiments; it pays in time rather than in trading losses. Prices recorded with extra measurement noise that is independent of the price are a garbling of the noiseless record, so by the data-processing inequality they carry less information, and the lower bound holds a fortiori.

The proof (Supplement S1) combines the Whittle form of the divergence rate with the resolvent identity $H_1 - H_0 = z^{-1} H_1 H_0 (G_1 - G_0)$: the two spectra differ by the leading unmatched moment, of order δ^2 for the pair and $\tau(\Delta\eta)^3$ for the moment-matched economies, and the divergence rate is the square of that difference. Numerically, the fitted slopes of the required sample length are -4.21 and -5.93, against the theoretical -4 and -6; for part (iii) the fitted slopes of the Whittle divergence rate in $\Delta\eta$ are 5.91, 9.98 and 14.05 for $K = 2, 3, 4$, against 6, 10 and 14.

Passive surveillance is possible in principle but needs at least the same polynomial order of observations as active probing needs experiments. Proposition S3.2 is the exact finite-sample version for the known pair.

6 What disclosure buys

Because active probing is costly and passive data are slow, the natural policy question is what a supervisor gains from disclosure. We compare six observation contracts at $\delta = 0.05$.

- I_0: public prices only.
- I_1: prices and aggregated executed flow.
- I_2: prices and one independent noisy report of dealer marked equity per reset ($\sigma_E = 0.5$).
- I_3: prices and exact audited equity (Proposition S24.3), also with reports rounded to 0.01.
- I_4: algorithm update timestamps without the update values.
- I_5: a supervisory audit registry with access to internal algorithm states.

Table 1: Disclosure contracts at $\delta = 0.05$ (pulse-design capital, par units)

Contract	Channel	Pulses	Benchmark	Saving vs. $T = 20$
I_0: prices, $T = 3$	public prices	271	4,305.25	−2,123.5%
I_0^*: prices, $T = 20$	full price path	12	193.62	0.0%
I_1: prices and signed flow	public flow	12	193.62	0.0%
I_2: noisy equity ($\sigma_E = 0.5$)	statement	12	193.62	0.0%
I_3: exact equity, 5-unit lot	audited equity	1	12.70	93.4%
I_3: exact equity, 0.01 rounding	rounded report	1	15.52	92.0%
I_4: timestamps only	no likelihood	12	193.62	0.0%
I_5: audit registry	internal states	0	0.00	100.0%

Closed-form values. Noisy equity adds information linearly in the number of resets, $D_{\mathrm{KL}} = N\Delta_E^2/(2\sigma_E^2)$, and saves only 1.5% against the $T = 3$ baseline. The equity difference is $\Delta E = \kappa^2 q \delta^2 V^2$; with reports rounded to 0.01, a guaranteed change needs $V \geq 5.59$.

Table 1 shows that disclosure is not one thing. Timestamps without update values carry no information about δ in this model. Noisy periodic statements barely help. Exact audited equity collapses the requirement to a single trade, limited only by lot sizes and report precision, and an audit registry removes the trading requirement altogether.

The cleanest form of useful disclosure is flow tagged by participant class.

Proposition 6.1 (Tagged flow removes the barrier)**.** *Let the price x_t be the causal stationary solution of the model driven by i.i.d. centred innovations of variance $\sigma_\xi^2 \in (0, \infty)$ in a positive, stable economy, with stationary autocovariances $\gamma_x(s) = \sum_\ell c_\ell \lambda_\ell^{|s|}$ ($c_\ell \geq 0$, $0 \leq \lambda_\ell < 1$ the closed-loop modes). Suppose cohort i has merged weight $w_i > 0$ and the observer receives prices and each cohort's flow $\tilde{u}_{i,t} = u_{i,t} + \varepsilon_{i,t}$, $u_{i,t} = q_i y_{i,t}$, with i.i.d. centred measurement noise of finite variance σ_f^2, independent of the whole innovation sequence and hence of the whole price path. Then $\tilde{u}_{i,t} = r_i \tilde{u}_{i,t-1} + w_i x_t + e_{i,t}$ with $e_{i,t} = \varepsilon_{i,t} - r_i \varepsilon_{i,t-1}$, and the just-identified IV estimator of $\beta_i = (r_i, w_i)$ with instruments $Z_t = (x_{t-1}, x_t)^\top$ satisfies $\sqrt{T}(\hat{\beta}_i - \beta_i) \to N(0, V_i)$,*

$$V_i = \Sigma_i^{-1} \Omega_i \Sigma_i^{-\top}, \qquad \Sigma_i = \begin{pmatrix} w_i A_0 & \gamma_x(1) \\ w_i A_1 & \gamma_x(0) \end{pmatrix}, \qquad \Omega_i = \sigma_f^2 \big[(1 + r_i^2)\Gamma_0 - r_i(\Gamma_1 + \Gamma_1^\top) \big], \quad (14)$$

where $A_m = \sum_{s \geq 0} r_i^s \gamma_x(s + m)$, $\Gamma_0 = \left(\begin{smallmatrix} \gamma_x(0) & \gamma_x(1) \\ \gamma_x(1) & \gamma_x(0) \end{smallmatrix}\right)$ and $\Gamma_1 = \mathbb{E}[Z_t Z_{t-1}^\top]$. The Jacobian determinant is

$$\det \Sigma_i = w_i \sum_{\ell,m} c_\ell c_m \frac{1 - \lambda_\ell \lambda_m}{1 - r_i \lambda_m} \; > \; 0. \quad (15)$$

V_i depends only on cohort i's own (w_i, r_i), σ_f^2 and the price autocovariances. Two conclusions are distinct: (a) for each admissible economy, $\hat{\beta}_i$ has the stated limiting normal law; (b) on a uniformly stable family with stationary price variance at least $\gamma_{\min} > 0$ and merged

*weights at least $w_{\min} > 0$, the limiting covariance V_i is bounded by a constant that does not depend on $|r_i - r_j|$. Statement (b) bounds the limit; it is not a central limit theorem that holds uniformly over the family. Three pieces are machine-checked: the finite-sample reset-experiment version, with an error bound that does not depend on the spacing; the lower bound on the determinant of the population instrument matrix that does not depend on the spacing (**`p1_tagged_iv_determinant_bound`**); and the exact decomposition of the IV score into reporting-error terms with coefficients fixed by the price path plus two boundary terms (**`p1_iv_boundary_decomposition`**). The stationary law of large numbers and the central limit theorem behind (a) are not machine-checked.*

In words. If each group's trading is reported separately, each group becomes an ordinary regression. Its precision depends on the group's own rate and on the price dynamics, but it has no singularity as the groups' rates approach each other.

Each tagged flow follows $u_{i,t} = r_i u_{i,t-1} + w_i x_t$, a regression whose information about w_i does not involve the other cohorts' rates; the closed form (15) matches the simulated Jacobian (0.1330 against 0.1322 in a two-cohort economy with 4×10^5 periods). In simulation ($T = 5{,}000$, 60 replications) the root-mean-square error of the recovered weights is flat in $\Delta\eta$ (fitted slope 0.011, error between 5.8×10^{-4} and 6.6×10^{-4}). This is the formal content of a disclosure regime such as a consolidated audit trail: tagging flow by participant class turns a polynomial barrier into an ordinary estimation problem.

6.1 How much disclosure is worth

Proposition 6.1 says that tagging removes the barrier; the next result says by how much.

Theorem 6.2 (Value of class-tagged disclosure)**.** *Take a rich class of Theorem 5.2: rates $r_c + \Delta\eta\, s$ with s in a compact set of γ-separated ordered points that has nonempty interior and contains the Gauss and Gauss–Radau pair of Theorem 5.1, and merged weights in a nondegenerate interval $[\omega_{\min}, \omega_{\max}]$, $0 < \omega_{\min} < \omega_{\max}$. In each reset experiment the observer applies a unit impulse and records the price response with independent $N(0, \sigma^2)$ noise per tap; under tagging it also records each class's flow response $f_{i,t} = w_i \sum_{s \le t} r_i^{t-s} h_s$ with independent $N(0, \sigma_f^2)$ noise per tap. Let N_p and N_t be the numbers of experiments needed, without and with tagging, for a fixed accuracy and confidence. There are constants $0 < c \le C < \infty$, depending on (σ, σ_f) and the class but not on $\Delta\eta$, such that*

(i) to recover the weights to a fixed accuracy smaller than half the weight difference between the Gauss and Gauss–Radau economies (so that recovering the weights decides between them), $c \le N_t \le C$, so $N_p / N_t \asymp (\Delta\eta)^{-(4K-2)}$;

(ii) to recover the rates to accuracy $c_0\Delta\eta$, for a fixed $c_0 > 0$ smaller than the radius of a ball contained in the class of rescaled nodes (so that no fixed guess meets the target), $c(\Delta\eta)^{-2} \le N_t \le C(\Delta\eta)^{-2}$, so $N_p/N_t \asymp (\Delta\eta)^{-(4K-4)}$.

Proof. $N_p \asymp \sigma^2(\Delta\eta)^{-(4K-2)}$ by Theorems 5.1 and 5.2. *Upper bounds.* Average the N price responses to estimate h with error of order $\sigma N^{-1/2}$ per tap. Given h, class i's mean flow is a two-parameter family $F(w_i, r_i; h)$ whose Jacobian columns $\sum_s r_i^{t-s} h_s$ and $w_i \sum_s (t-s) r_i^{t-s-1} h_s$ are linearly independent, with a Gram determinant bounded below uniformly on the class because $w_i \ge \omega_{\min}$, the rates lie in a compact subset of $(0, 1)$ and $h_0 = \kappa > 0$; the bound does not involve the other classes or $\Delta\eta$. Least squares on class i's flows with the plug-in h therefore estimates (w_i, r_i) with error $O_P(N^{-1/2})$, uniformly on the class, which gives the upper bounds in (i) and in (ii) for accuracy $\epsilon\Delta\eta$. For (i) there is also a finite-sample version: in the first period the tagged flow of class i is $w_i\kappa$, so the mean of the N first-period records gives $P(|\hat{w}_i - w_i| \ge \epsilon) \le \sigma_f^2/(N\kappa^2\epsilon^2)$ by Chebyshev's inequality, for every configuration (this step is machine-checked). *Lower bounds.* At a configuration in the interior of the class, perturb one class's weight by ϵ, or its rate by $\epsilon\Delta\eta$; for small ϵ both perturbed configurations remain in the class because the weight interval is nondegenerate and the rate set has nonempty interior. The mean differences of both the price and the flow records are of order ϵ, respectively $\epsilon\Delta\eta$, so the Kullback–Leibler divergence of N experiments is at most $CN\epsilon^2$, respectively $CN\epsilon^2(\Delta\eta)^2$, and Le Cam's two-point lemma gives the lower bounds. □

In words. This is the value of disclosure measured in data. Tagging replaces a requirement that explodes as the groups become similar with one that does not grow at all (for the weights) or grows only like $(\Delta\eta)^{-2}$ (for the rates).

Proposition 6.1 and Theorem 6.2 use different observation arrangements: the proposition concerns passive, stationary data with prices observed without noise, where the price is a valid instrument; the theorem concerns reset experiments with noisy prices. With noisy prices, regressing flow on the observed price is an errors-in-variables problem, which is why the theorem works from the known impulse instead.

Proposition 6.3 (Conditional limits of three evasion arguments)**.** *Maintain the exogenous reporting-noise and valid-instrument assumptions of Proposition 6.1; fix the finite number of linked class labels and a uniformly stable model family.*

(i) For a fixed split $u_{i,m} = \alpha_{i,m} u_i$ with known common ownership, independent mean-zero reporting errors and $\alpha_{i,m} \ge \underline{\alpha} > 0$, only the lagged-flow column of the population IV matrix scales. Thus $\Sigma_{i,m} = \Sigma_i \operatorname{diag}(\alpha_{i,m}, 1)$ and $\det \Sigma_{i,m} = \alpha_{i,m} \det \Sigma_i$. Subject to the remaining uniform moment and excitation conditions, fixed finitely many consistent

sub-account estimates can be summed: the true identity is $w_i = \sum_m w_{i,m}$*, and the estimated sum has sampling error. The share restriction implies* $M_i\underline{\alpha} \leq 1$*. This is not robustness to arbitrarily many vanishing accounts; a common estimated rate does not identify ownership.*

(ii) Assume separately an execution cost lower bound $\mathbb{E}C_\zeta \geq c_\zeta \sigma_\zeta^2$ *with* $c_\zeta > 0$ *and a gross profit upper bound* $\mathbb{E}\Pi \leq \overline{\Pi}$ *uniform in the spacing. Camouflage is unprofitable when* $c_\zeta \sigma_\zeta^2 > \overline{\Pi}$*. A variance requirement of order* $(\Delta\eta)^{-(4K-2)}$ *requires an additional additive exogenous-noise estimation model with uniformly controlled instruments. Neither this variance requirement nor the execution bound is established for arbitrary strategic orders.*

(iii) If intensity is scaled to $\gamma_i w_i$ *while* $1 - r_i \geq \eta_{\min} > 0$*, then* $q_i \leq \gamma_i w_i / \eta_{\min} \to 0$ *as* $\gamma_i \to 0$*. The other groups remain in the aggregate budget; reducing one contribution does not itself establish that the ceiling is met.*

Economic scope. These are conditional calculations about fixed linked accounts, bounded execution costs and a persistence class. They do not solve an equilibrium game of evasion. In particular, endogenous orders can affect prices and invalidate the reporting-error instrument, and a persistence change can offset a reduction in weight. Adversarial account creation and response adaptation require further modelling.

In a model calculation using the same assumed noise and lags for both observers, the Cramér–Rao requirements give fitted slopes of -6.08 for prices and -4.34 for the ratio at $K = 2$ (theory -6 and -4). Calibrated to the descriptive persistence estimates below, prices alone would need about 333 times as many days as the published class ratios to separate the top-trader and all-account memories in BTCUSDT (rates 0.9889 and 0.9986 per five minutes; about 6,000 days against 18), and about 19 times as many in ETHUSDT (rates 0.9793 and 0.9972). These rates come from day-demeaned regressions, which bias persistence down (Supplement S19), so the factors are illustrative.

6.2 When disclosure is worth acquiring

When is disclosure worth its cost? In a stylised cost comparison (Proposition S27.1, Supplement S27), a supervisor prefers an audit trail when the harm from slow attribution exceeds the trail's cost per market. Disclosure is most valuable in a middle band of memory spacings, where prices are too slow and labels are fast, and only when the trail covers enough markets to share its fixed cost.

A decision in which attribution matters. A supervisor chooses between a uniform reduction in activity and halting the largest verified class, over a declared supervisory horizon. The target below is the feedback budget $S = a + \kappa \sum_i q_i$, not the spectral radius of the state matrix. In an already stable positive economy, imposing $S \le \bar{s} < 1$ is a tighter prudential constraint, not restoring stability to an unstable market.

Proposition 6.4 (Supervisory separation: market-wide versus targeted intervention)**.** *Assume the positive stable model, known a, κ, and $a < \bar{s} < 1$. Write $q_i = w_i/(1 - r_i)$ and $R = S - \kappa \max_i q_i$ for the post-halt feedback budget.*

(i) A uniform scale θ meets the ceiling exactly when $a + \theta(S - a) \le \bar{s}$. Its largest feasible value is $\theta^* = \min\{1, (\bar{s} - a)/(S - a)\}$. The identity $S = 1 - \kappa/\sum_j h_j$ is algebraic; estimating this infinite sum requires a justified tail bound and an observation/noise model. It alone establishes no unconditional parametric learning rate.

(ii) For $K = 2$, the Gauss and Gauss–Radau pair of Theorem 5.1 has limiting largest normalized weights $1/2$ and $3/4$. Thus $|S^G - S^R| = O((\Delta\eta)^3)$ and $q^R_{\max} - q^G_{\max} = W/(4(1 - r_c)) + O(\Delta\eta)$. For sufficiently small spacing, a nonempty interval of ceilings separates their post-halt decisions. For N independent reset records with common Gaussian noise $\sigma^2 I$, both decision errors at most $1/4$ require

$$N \ge (2\Phi^{-1}(0.75)\sigma/\|h^G - h^R\|_2)^2 \ge c\sigma^2(\Delta\eta)^{-6}.$$

The record horizon and the response norm must be the same on both sides.

(iii) Tagged-flow estimation under Theorem 6.2, a rate bound away from one, and a fixed nonzero decision margin gives a bounded sample requirement as the spacing shrinks. The target class must be mapped to an executable, verified class label; a largest latent atom alone supplies no such identity.

Proof. Part (i) follows by scaling the intensities. For (ii), expand $(1 - r_c - \Delta\eta s)^{-1}$: the first three moments agree, so aggregate budgets differ by order $(\Delta\eta)^3$. Each intensity converges to $W\omega_i/(1-r_c)$, giving the stated dominant-weight gap. This exceeds the aggregate discrepancy at sufficiently small spacing. A ceiling between the two post-halt budgets reduces the action problem to testing two Gaussian mean vectors. Their common-covariance testing error is $\Phi(-\sqrt{N}\|h^G - h^R\|_2/(2\sigma))$; the response bound in Theorem 5.1 gives the lower bound. Part (iii) follows from continuity of $w/(1 - r)$ on the declared compact parameter set and the fixed margin. □

General-K scope. The same reduction gives exponent $4K-2$ only for a pair whose price responses have that moment-matching order *and* whose dominant intensities have a separately established fixed gap. Moment matching alone does not supply the latter. In particular, the symmetric odd/even binomial pair has equal maximal normalized weights. The $K=3$ rows below are numerical examples, not a construction proving this condition for every K.

Formal verification scope, 5 November 2026. This revision builds the pinned Lean 4.34.1/Mathlib project and audits its typed declarations using only standard foundations. The exact two-point Gauss–Radau witness in Table S36 is now covered by `Paper1GaussRadau.lean`: positive normalized weights, moments of orders zero through two, the third-moment gap, a dominant-intensity gap of at least $W/(8u)$ for $0 \le \Delta\eta \le u/3$, the aggregate-budget gap and the actual finite-Gaussian-record testing bound. Thus this example no longer relies solely on the other matched pair used by the earlier supervisory theorem. The finite-horizon bound has constants depending on the declared record length and response envelope. Its proof does not supply an all-K quadrature construction. `Paper1SupervisoryDecision.lean` additionally checks the robust threshold/fallback, the actual unsafe-halt event for Proposition 6.5, the fixed-menu no-data optimum and the acquisition algebra of Proposition 6.6. The new receipt binds the build to hashes of the active Paper 1 sources. Declaration counts include supporting results across the project and are not counts of fully formalized manuscript statements. Independent human review, novelty and empirical market assumptions are not certified by Lean.

Formal verification additions, 6 November 2026. Pinsker's inequality, which Mathlib lacks, is now proved in Lean in the form the paper uses (`Paper1Pinsker.lean`): $2(P(A)-Q(A))^2 \le D_{\mathrm{KL}}(P\|Q)$ for every measurable event A. With it the testing step holds at every error level, and Lean checks the following in the paper's own form (`Paper1GeneralLevel.lean`): Theorems 5.1 and S25.1 on a finite horizon, where any test with error probabilities α and β needs $4\sigma^2(1-\alpha-\beta)^2 \le NT(\varepsilon C_T)^2$; Theorem 4.1(i) as stated, a worst error of at least $\frac{1}{2}(1-K_*V\delta^2/(2\sigma))$ and hence $V \ge 2\sigma(1-2\alpha)/(K_*\delta^2)$; and the capital bound (S18) of Lemma S24.4, with a formal margin constant $m>0$ of the paper's form. Theorem S3.1 is machine-checked end to end for rules given by measurable sets of paths (`Paper1Abstain.lean`): the autoregressive path laws are unit-triangular images of i.i.d. Gaussians, the divergence between two of them is at most $(s_1-s_0)^2MT(T-1)/4$ when $s_0^{2k} \le M$, and both the abstention bound and the binary-rule bound follow as stated; randomized rules are not covered. Corollary 5.3(i)–(ii) is machine-checked (`Paper1UnknownPrimitives.lean`): if two

economies with unknown and possibly different (a, κ) have the same price response and nonzero memory rates, they have the same κ, the same a and the same merged measure; part (iii) is proved on paper.

Formal verification additions, 7 November 2026. Proposition S7.1(i) is machine-checked on the raw records (`Paper1GaussianHCR.lean`, `Paper1GaussianCR.lean`, `Paper1CRActual.lean`, `Paper1CRRaw.lean`). The Gaussian likelihood ratio and its second moment are derived from the densities. The Hammersley–Chapman–Robbins inequality then holds for every square-integrable statistic. A limit along a line gives $\mathrm{Var}_\theta(\hat{w}_k) \geq \sigma_h^2 [F_{\mathrm{unit}}^{-1}]_{kk}$ for every statistic of all N raw reset records that is unbiased near θ. Here J is the true Jacobian of the actual closed-loop response; Lean proves that this response is differentiable in the cohort sizes and rates. Part (iii) is machine-checked for both weight coordinates on the stated family (`Paper1FisherBounds.lean`, `Paper1CRTaps.lean`, `Paper1CRS71iii.lean`, `Paper1CRS71iiiEta.lean`, `Paper1CRS71iiiW2.lean`). The taps are those of the actual closed-loop recursion and J is their true Jacobian. The lower bound uses the direction $v = (1, -1, \Delta\eta/(2w_1), \Delta\eta/(2w_2))$, along which every kernel moment moves by $(\Delta\eta)^3 Q_m$ with $|Q_m| \leq m^3$, so $\|Jv\|^2 \leq B_T(\Delta\eta)^6$. The upper bound uses an explicit combination of $h_1, \ldots, h_4$ that recovers w_k to first order, built from the inverse row of the confluent Vandermonde Jacobian. Switching between r_i and $\eta_i = 1 - r_i$ leaves the weight entries unchanged. Theorem 5.4(ii) is machine-checked for every K, starting from equal spectra at every frequency (`Paper1SpectralID.lean`, `Paper1SpectralH.lean`). Two stable economies satisfying (i), with possibly different numbers of cohorts, anchors, impacts and innovation variances, have equal spectra only if they have equal $\sigma_\xi^2 \kappa^2$, equal rate polynomials $\prod_i (z - r_i)$ (hence the same rates) and equal denominators of H. With κ known they also have the same anchor and the same merged weights. Stability enters as the condition that the poles of H lie in the open unit disk. This replaces the earlier two-cohort result, which started from equal numerator polynomials rather than equal spectra. Proposition S3.2 is machine-checked end to end for the explicit block model (`Paper1PassiveVariance.lean`). This covers the block mean and second moment from the innovation and noise moments, Var $\leq R$ when $\mathbb{E}w^4 \leq 3s_w^4$, Chebyshev for n independent blocks, and both midpoint errors at most α once $n \geq 4R/(\alpha\Delta^2)$. The converse in Lemma S24.4 is machine-checked (`Paper1CapitalConverse.lean`): the error probabilities of the single-pulse test on the third observation, and margin solvency of the funded pulse trader at every later mark. Parts (ii) of Propositions S5.1 and S5.2 are machine-checked (`Paper1KalmanTied.lean`). In Proposition S5.2, a bound on the second derivative of each tap function $w(\eta)(1-\eta)^m$ is a stated smoothness hypothesis, and part (iii), which uses Brouwer's theorem, is not machine-

checked. Still written-only: the stationary IV central limit theorem of Proposition 6.1, and the divergence-rate form and the Θ bounds of Theorem 5.5.

Explicit robustness condition. Smooth best responses alone do not preserve separation: the constant smooth map $\Phi(M) = 0$ sends both post-halt budgets to a. Instead assume that, over the stated action and holding horizon, $|\widetilde{R}_j - R_j| \le \varepsilon$ for $j \in \{G, R\}$, with $R_R < R_G$. Then

$$R_R + \varepsilon \le \bar{s} < R_G - \varepsilon \quad \Longrightarrow \quad \widetilde{R}_R \le \bar{s} < \widetilde{R}_G. \tag{16}$$

Indeed, $\widetilde{R}_R \le R_R + \varepsilon$ and $\widetilde{R}_G \ge R_G - \varepsilon$. A nonempty robust window exists if $2\varepsilon < R_G - R_R$. This is a maintained response bound, not an equilibrium result. The same allowance must cover the alternative uniform action. At the $K = 2$, $\Delta\eta = 0.10$ calibration, $R_R = 0.532$, $R_G = 0.569909$, and $\bar{s} = 0.55$ has margin 0.018; an allowance 0.005 fits within it.

A complete conditional cost-and-error model. Let j denote the economy, π_j its declared prior, H the holding horizon, b the surplus per active class per period, C_h the implementation cost of halting and L the penalty per period for violating the ceiling. These are assumed accounting primitives, not estimated social welfare. An envelope $S_{\max}$ supports the robust uniform action

$$\theta_\varepsilon = \min\{1, \max\{0, (\bar{s} - \varepsilon - a)/(S_{\max} - a)\}\}, \qquad L_T = HbK(1 - \theta_\varepsilon).$$

This action is feasible only if its worst-case budget meets the ceiling (in particular $a + \varepsilon \le \bar{s}$). For a halt,

$$L_{h,j} = Hb + C_h + HL\,\mathbf{1}\{R_j + d_j > \bar{s}\}, \quad \mathcal{L} = F + Nc + \sum_j \pi_j\{p_j L_{h,j} + (1 - p_j)L_T\}.$$

Here $F + Nc$ is the information bill, p_j the probability of halting, and $|d_j| \le \varepsilon$ the action response. Require $\Pr_j(\text{halt and violate}) \le \alpha$ separately in each world. Compare this loss with the best feasible no-purchase lottery over the *same two actions*, and allow refusing to buy information. Neither arbitrary alternative policies nor an optimum over sample size are claimed. Active probing additionally incurs its execution loss and capital requirement; these are not automatically the per-record fee c and must not be counted twice.

Existing runs and the remaining synthetic check. The archived `R_P1_targeted_intervention.json` supplies six response/decision rows. Recomputing them independently from the stated state recursion agrees to 2.1×10^{-17} in the response discrepancy. Those

rows supply S_j, R_j and Gaussian testing distances; they contain no measured disclosure fees, intervention payoffs or strategic response bounds. The existing GPU and historical diagnostics cannot fill these quantities by relabelling their outputs. The new CPU study `studies/supervisory_release_20261105/run_study.py` therefore evaluates only this missing accounting layer, with exact Gaussian probabilities rather than a new large simulation.

For this check alone, assume independent scalar disclosures $Z_n \sim N(R_j, \sigma_t^2)$ and halt when $\bar{Z} + z_{1-\alpha}\sigma_t/\sqrt{N} + \varepsilon \le \bar{s}$. This is an illustrative measurement contract, *not* a claim that class positions or tagged flow directly observe post-halt budgets. If a world is unsafe under the response allowance, this gate falsely halts with probability at most α under the stated unbiased Gaussian law. The price comparator observes the full 400-tap Gaussian reset record of the same known pair, with a likelihood-ratio rule constrained by the same per-world error budget. Both information channels have the same assumed monetary fee here; neither represents free passive observation.

With $K = 2$, $N = 32$, $\sigma_t = 0.02$, $\varepsilon = 0.005$, $\alpha = 0.05$, $\pi_R = 1/2$, $H = b = 1$, $C_h = 0.02$, $L = 10$, $F = 0.05$, and $c = 0.0005$, the uniform loss is 1.449219, the information bill is 0.066, and disclosure loss is 1.305132. Its conditional net advantage is 0.144087 surplus units. Halt probabilities are approximately zero in G and 0.978928 in R. Buying the price record yields loss 1.515219 at this sample size; its optimal acquisition decision is to decline and retain the no-data action. An informed oracle before information fees has loss 1.234609.

Across 2,304 deterministic configurations varying fees, noise, horizon, prior, sample size and allowance, paid disclosure improves on the no-data benchmark in 1,207, worsens it in 1,079, and ties in 18. These counts have no population or statistical-significance interpretation. Negative cases remain in the output. A downward measurement bias of 0.03 produces false-halt probability 0.418776; a future response of 0.025 despite an allowance of 0.005 makes halting unsafe in R with probability 0.978928. The latter also invalidates the uniform-action guarantee, so payoff comparisons under that misspecification are not presented as a fair safety comparison. The experiment establishes conditional accounting and failure modes, not real-market gains, a validated equilibrium response, or the general-K theorem.

Proposition 6.5 (A robust scalar disclosure gate)**.** *Let $Z_1, \dots, Z_N$ be independent $N(R, v)$ readings of the fixed post-halt feedback budget, $N > 0$, and let the realized action response d satisfy $d \le \varepsilon$. For $t > 0$, halt only if $\bar{Z} + t + \varepsilon \le \bar{s}$. Then the probability of halting while the realized budget exceeds the ceiling is at most*

$$\Pr\{\bar{Z} + t + \varepsilon \le \bar{s},\ R + d > \bar{s}\} \le \frac{v}{Nt^2}.$$

Consequently $v \le \alpha N t^2$ gives a per-world error bound α.

Proof. On an unsafe-halt event, $\bar{Z} + t + \varepsilon \le \bar{s} < R + d \le R + \varepsilon$, so $t \le |\bar{Z} - R|$. Chebyshev's inequality for the actual Gaussian sample mean gives the bound. Lean checks the product Gaussian law, the sample statistic, this event inclusion and the probability inequality in `p1_supervisory_gaussian_unsafe_halt`; it is not merely a check of a scalar named "probability bound." The response allowance and the interpretation of Z remain explicit assumptions. □

This conservative gate uses $t = \sigma_t/\sqrt{N\alpha}$. It differs from the sharper Gaussian-quantile gate above, whose exact normal probabilities are numerically evaluated but whose quantile identity is not formalized here. Under the same calibration with $N = 128$, the conservative gate has radius 0.007906, information bill 0.114, loss 1.349034 and conditional net advantage 0.100185. Its halt probabilities are approximately zero in G and 0.998023 in R. Across the same 2,304 sensitivity configurations it improves on the fixed-menu no-data benchmark in 781, worsens it in 1,499 and ties in 24. Both gates' rows remain in the result file: 4,608 configurations in total. The welfare and error interpretation remains conditional on the declared observation and response contracts.

Operational bridge from tagged flows to a halt decision. The scalar-reading contract above can be replaced, within the reset model, by five observed channels: two first class-flow responses, two second responses, and the first price response. For a unit pulse write $f_{i,0} = \kappa w_i$, $f_{i,1} = w_i(\kappa r_i + h_1)$, and $h_1 = \kappa a + \kappa^2 \sum_i w_i$. Thus

$$w_i = f_{i,0}/\kappa, \qquad r_i = f_{i,1}/f_{i,0} - h_1/\kappa, \qquad q_i = w_i/(1 - r_i).$$

Assume known impact, verified stable class membership, $0 \le r_i \le 0.95$, independent unbiased Gaussian reset readings with per-channel variance σ_f^2, and N independent records. A simultaneous radius $e = \sigma_f\sqrt{5/(N\alpha)}$ covers the five sample means with probability at least $1 - \alpha$ by Chebyshev and a union bound. Interval division, only when the first-flow lower endpoint is positive, gives bounds on w_i, r_i, q_i; intersect rate bounds with the declared class. Halt only if the remaining upper budget plus the response allowance meets the ceiling. Otherwise retain the same robust uniform fallback. On simultaneous coverage, every selected halt is safe. This supplies a conditional observation-to-action argument without feeding the unknown R, w or r to the rule.

The separate Lean module `Paper1FlowDecision.lean` covers exact flow inversion, ratio and clipped intervals, the intensity interval, selected-halt inequality, actual product-Gaussian joint mean bound and a probability-composition theorem with an explicit rule-soundness

premise. Their deterministic composition, clipping implementation and executable simulation still require separate correspondence checks; no claim of a fully formalized executable decision algorithm is made. The pinned build receipt records the checked declarations.

The exploratory script `studies/paper1_uplift_20261105/flow_decision.py` runs 20,000 independent replications per world in each of 60 configurations: $N \in \{32, 128, 512, 2048, 8192\}$, $\sigma_f \in \{0.0005, 0.002, 0.005\}$, $H \in \{1, 10\}$ and $F \in \{0.05, 0.25\}$. Other quantities use the two-group calibration above, a per-channel-vector record fee 0.0005, and the same class-halt or throttle action menu. Fourteen configurations have positive estimated net information value; the others are retained. The largest observed gain is exploratory, selected from the grid: at $N = 128$, $\sigma_f = 0.0005$, $H = 10$ and $F = 0.05$, no-data loss is 14.492188, information-inclusive loss 12.370094 and net gain 2.122094 surplus units. A simultaneous grid-adjusted Hoeffding interval is reported in the result file; zero observed unsafe halts is not a proof of zero probability. Costs, horizons and response bounds are synthetic, and no real-market gain or calibrated welfare conclusion follows. Six additional fixed stress cases retain both smaller biases with no observed unsafe halt and failures: a downward second-flow bias of 0.010 produces unsafe halts in G in all 20,000 replications, while a response 0.030 exceeding the allowance 0.005 produces unsafe halts in R in all replications. Swapped second-flow labels lead to abstention in both worlds. These are synthetic contract falsifications, not guarantees under misspecification.

Observation assumptions in market data. The required class records are signed executed quantities with stable verified labels, synchronized timing, known probe history and a justified error model. Public position ratios and weekly position stocks do not satisfy this contract. Binance's top-position ratio concerns the top 20 percent of users by margin balance; the top-account ratio counts accounts once. Their populations can change and overlap with the all-account population, so those series are not disjoint cohorts or algorithm labels. CFTC reports describe Tuesday positions generally published on Friday, and business-purpose categories can contain different trading motives. Provider definitions are available at https://www.cftc.gov/MarketReports/CommitmentsofTraders/index.htm and https://developers.binance.info/en/docs/catalog/core-trading-derivatives-trading-usd-s-m-futures/api/rest-api/market-data. Neither source identifies a learning update rule. Their fitted persistence and sample-size ratios remain descriptive diagnostics. Converting stock persistence to flow response also requires holdings, valuation, transfers, liquidations, reporting delay and changing-membership adjustments; a stock–flow identity alone supplies no causal identification. Real-market use requires verified event-level class records or a justified obser-

vation model, external excitation or valid instruments, and a separately tested post-action response allowance.

Proposition 6.6 (No-data action and information purchase)**.** *For the fixed halt/throttle menu, let* $\overline{L}_h = \sum_j \pi_j L_{h,j}$ *and let* $p_{\max} = \alpha$ *if a halt is unsafe in at least one world, otherwise* $p_{\max} = 1$*. Assuming the fallback meets the ceiling, the best constant no-data halt probability is* $p = 0$ *when* $\overline{L}_h \geq L_T$*, and* $p = p_{\max}$ *when* $\overline{L}_h < L_T$*. A specified information rule with expected action loss* L_I *is strictly preferred if and only if*

$$F + Nc < \mathcal{L}_0 - L_I.$$

Allowing refusal to purchase gives loss $\min\{\mathcal{L}_0, F + Nc + L_I\} \leq \mathcal{L}_0$.

Proof. The no-data loss is affine in its feasible halt probability: $L_T + p(\overline{L}_h - L_T)$. Its slope determines the minimizing endpoint. Subtracting the information rule's total loss gives the break-even inequality, and the minimum inequality proves the optional-purchase statement. These algebraic decisions are machine checked in `Paper1SupervisoryDecision.lean`. They do not establish an optimum over other actions, sample sizes, or information contracts. □

What the data can and cannot test. The theorems concern experiments that no outside researcher can run: reset initial states, known price persistence and impact, and either controlled probes or stationary data from a known innovation law. Public data meet none of these exactly. Order flow is endogenous, the fundamental value is unobserved and replaced by returns, observations overlap in time, parameters drift, and the class reports are positions aggregated over many accounts rather than tagged flow. The tests below therefore check the theory's observable prediction, that class records restrict persistence which prices leave unrestricted, under a stated mapping from class positions to cohort states. They do not estimate the theorems' rates, and a pass does not show that class persistence is learning memory.

6.3 Evidence from trader-class reports

Two public data sets report positions by trader class: the CFTC's weekly Commitments of Traders, mandated by law, in twenty regulated futures markets, and Binance's five-minute top-trader and all-account ratios, published voluntarily, in 30 perpetual markets. Supplements S18 and S19 report the pre-specified tests in full; the formal two-root stock–flow inventory bridge linking Remark 2.2 to reported position stocks and flows is derived at the

beginning of Supplement S18. In both datasets, class positions pin each class's fitted persistence to a narrow interval while prices leave it essentially unrestricted: the primary CFTC hypothesis holds in all eight markets, with price-only intervals 9 to 71 times wider, and in all twelve markets of a later registered replication; matching the weekly report's precision from prices alone would take, at the median, about 1,350 times as much data under standard root-T scaling (a hypothetical conversion, not an identified data requirement; Table S26). On 30 fresh Binance markets the horizon-matched comparison gives a median width ratio of 7.3. Several pre-specified tests failed and are reported as they came out (Table S29): Binance class data did not improve out-of-sample return forecasts, the CFTC reports read as flows pinned down their own persistence in only 3 of 8 markets (6 were required), and a test designed to separate learning memory from position turnover found a distinct faster root in only 2 of 8. These failures are consistent with several explanations, including within-class sign cancellation: coarse reporting buckets (e.g., CFTC Leveraged Funds) mix offsetting strategies (such as cash–futures basis arbitrageurs trading against trend-following momentum CTAs), motivating tests of granular algorithm-tagged records without proving their necessity.

7 Implications for supervision and market design

The results separate two supervisory tasks that are often run together. Monitoring aggregate fragility, for example to set market-wide limits, needs only the combined price response, which public data can identify under the observation and tail conditions of the model (Remark S25.2, Proposition 6.4(i)). Attributing feedback to particular groups, to apply targeted limits or enforcement, faces the $(\Delta\eta)^{-(4K-2)}$ barrier whenever groups react over similar horizons (Proposition 6.4(ii)). Class-tagged records remove that barrier group by group (Proposition 6.1), and in the model persistent pseudonymous class labels deliver the full statistical gain without personal data.

What a supervisor can learn from public data. Prices and anonymous flow reveal the merged measure M and the closed-loop impulse response, hence the transfer function, and nothing finer. A supervisor who wants to know *which* algorithms drive a feedback loop cannot get there from public data at reasonable cost when their memories are close. This is not a failure of econometric technique: Theorem S25.1 holds for every estimator, biased or not.

What probing costs. At realistic observation noise, an exchange or a large desk could separate two well-separated memory rates for about 10^5 USDT of capital. Probing is therefore feasible for coarse questions and infeasible for fine ones. Because capital grows as δ^{-4} and experiments as $(\Delta\eta)^{-(4K-2)}$, halving the separation multiplies the cost by 16 for a known pair and by 2^{4K-2} for K unknown cohorts.

What disclosure buys. Under the model's stated observation contracts, verified class-tagged executed flow can remove the spacing penalty for fixed-accuracy weight recovery (Proposition 6.1). The attribution experiment compares those records with specified periodic statements; its ranking does not apply to every possible disclosure design. The factor of about 333 obtained from Binance's day-demeaned ratio persistences is an illustrative model calibration. Top traders and all accounts overlap, and the published ratios are not the disjoint executed-flow records used by the theorem. CFTC position reports likewise provide descriptive information about fitted persistence, without validating the theorem's observation contract or identifying algorithmic memory.

When audit trails already exist. The US Consolidated Audit Trail (SEC Rule 613) is designed to capture specified order and reportable-event information for NMS stocks and listed options across exchanges and broker-dealers, from origination to execution or cancellation, linked to a customer identifier and to the broker-dealer that handled the order; its actual scope is governed by Rule 613, the CAT NMS Plan and exemptions. Since February 2025, under SEC exemptive relief, it no longer requires the names, addresses and years of birth of natural-person customers, who remain linked through an identifier derived from a transformed tax identifier (SEC, 2012, 2016, 2024, 2025). Our results bear on such regimes in three conditional ways. First, for the attribution task modelled here, records that report each class's flow separately remove a barrier that, for an observer who learns by probing, costs round-trip trading loss of order δ^{-4}, and they save samples by a factor of order $(\Delta\eta)^{-(4K-2)}$ at fixed accuracy or $(\Delta\eta)^{-(4K-4)}$ for rates resolved to a fraction of their spacing (Proposition 6.1, Theorem 6.2). This is a measurement value, not a cost–benefit assessment: CAT's budget pays for enforcement, market reconstruction and surveillance, none of which the model measures, and Supplement S8 sets the hypothetical probing loss beside it only as a scale illustration. For estimating class responses in our model a persistent pseudonymous label suffices; this does not settle what enforcement or data-minimization design requires.

Second, an audit trail may connect accounts, entities and orders without identifying a homogeneous decision rule. Within-class heterogeneity and opposite-signed responses can produce cancellation, but the reported regression failures do not identify cancellation as their

cause. Finer strategy or algorithm labels could help only if membership, timing, executed quantities and the reporting-error model meet the identification assumptions. An algorithm identifier does not by itself verify a fixed linear response, ownership linkage or persistence. The paper therefore establishes neither that a particular reporting granularity is necessary for every supervisory task nor that finer identifiers automatically restore a parametric learning guarantee.

Third, the audit trail does not reach the markets we study. Crypto perpetual futures are outside it, futures regulated by the CFTC have separate reporting, and its data are not public: regulators and authorized self-regulatory organizations have controlled access, while outside desks and researchers generally do not. For these observers the price-only and coarse-disclosure contracts of this paper are the relevant ones.

What to monitor instead. Because transient loss can be arbitrarily large at fixed eigenvalues and volatility (Theorem S12.1), supervision that wants to control risk rather than identify participants should monitor quantities that bound transient loss directly: the summed absolute impulse response and confidence sets for the transfer function, both of which are identified from public data. These are cheap where identification is expensive (Remark S25.2); the companion paper shows that the summed response can be bounded at the parametric rate from controlled experiments, and only under a declared bound on memory length from a finite window (Niroula et al., 2026).

How to measure feedback. Empirical feedback indices should (i) compare persistence with the mechanical value of the filter, (ii) align flow with the bar in which it is observed, (iii) control for order-splitting persistence or use instruments with a defended exclusion, and (iv) report placebo pipelines. Each of these changes the sign or the order of magnitude of published-style estimates in our data.

8 Conclusion and an open problem

In a linear market with exponential-memory cohorts and known impact parameters, prices reveal a merged summary of algorithmic feedback. Resolving the model cohorts costs executed quantity of order δ^{-2} and trading capital of order δ^{-4} for a known pair, with a numerically evaluated constant of about 61 par units at the calibration, and a number of experiments of exact order $(\Delta\eta)^{-(4K-2)}$ for K unknown cohorts, and at least that order of passive observations; in simulations of stylized nonlinear rules and of online learners trained in the closed loop the same costs hold when the competing populations' feedback weights

match, populations whose learned weights differ are told apart by those weights instead, and hard thresholds hide cohorts from small probes altogether. Without the exponential form, accuracy improves only as $1/\log N$, with matching lower and upper bounds. Class-tagged disclosure removes the barrier under the stated contract, and its value can be stated in samples: a factor of order $(\Delta\eta)^{-(4K-4)}$ for resolving rates (and $(\Delta\eta)^{-(4K-2)}$ at fixed accuracy), an illustrative model-calibration factor of about 333 using the published BTCUSDT ratio persistences, without establishing that the actual reporting series meet the theorem's observation contract. The CFTC's class reports restrict a fitted position persistence that prices leave unrestricted in all eight regulated markets we examined; this persistence is not shown to be learning memory, a pre-specified test of what prices do reveal failed, and class-tagged memory states did not improve 30-minute post-shock return forecasts ($\Delta R^2 = -1.09$ points, inconclusive). Differencing weekly CFTC position stocks under an additional stock–flow interpretation gives descriptive response regressions, which failed a pre-specified precision test, and a pre-specified test for a second, faster root in positions, which would have been the learning memory, failed as well. With real Bitcoin price noise in place of Gaussian noise, the predicted rates held in a pre-specified test, at the sample sizes where the real noise could be used directly. Under the specified Roll-noise calibration the model capital bound for a moderate separation is about 10^5 USDT; this is not a measured real-market identification bill. Estimated feedback associations in crypto markets are small, rises with depth, does not move when fees change trading volume severalfold, and is easily confused with filtering and timing artefacts. Finally, eigenvalues and volatility cannot certify safety, because transient loss can be arbitrarily large at fixed spectrum and variance.

Two open questions remain. The first is economic: Proposition 6.4 gives one stylised decision in which knowing the individual groups changes the right action, but turning the information costs here into a value of disclosure needs a supervisory loss function estimated from real interventions. The second is for supervision: *what can be measured and controlled without identifying the algorithms?* The impulse response and the transfer function, and finite-sample confidence sets for them, are identified from public data even when cohorts are not. The structural matrix J is not: economies with the same merged measure can have different state dimensions, so J is identified only up to the equivalence class of minimal realizations of the same transfer function. A well-posed next question is: for a stated class of stable closed-loop markets, given an estimated transfer function and a finite-sample confidence set, which intervention policy of least cost guarantees a stated bound on future price displacement without recovering individual cohorts? The companion paper uses exactly this interface to build an identification-free pre-trade control.

Declarations

Funding. This research received no external funding.

Competing interests. The authors declare no competing interests.

Author contributions. Pratik Niroula: conceptualization, methodology, software, formal analysis, writing (original draft). Proposed review responsibilities for confirmation: Prashant Niroula and Sujan Khatiwada, proof review, validation and manuscript review. Independent human proof review and final approval by each author remain to be recorded; this draft does not attest that they have occurred.

Data availability. All data are public. Binance market data come from the Binance public data archive (`https://data.binance.vision`): spot and USD-margined perpetual futures candles at one-second and one-minute frequency, futures order-book depth, and the published trader-class long/short ratios (top traders by position and by account, and all accounts); every file is verified against Binance's published checksums. The CFTC Commitments of Traders reports (Traders in Financial Futures and Disaggregated, futures only, through 13 October 2026) come from the CFTC public reporting API. Weekly prices for the CFTC markets (stock index, currencies, Treasury yield and CME futures) come from Yahoo Finance. Processed data and all result files are in the public repository below.

Code availability. All code, pre-specified test specifications, result files and test-unlock counters are public at `https://github.com/Pratikn03/Fianince`. The repository's README lists the scripts in run order, and Supplement S32 maps every number in the paper to the file that produces it.

Ethics approval. Not applicable. The study uses only public market data and involves no human participants or personal data.

Use of AI tools. Claude (Anthropic) assisted with data analysis, analysis code, derivations, numerical results and text. Gemini (Google) assisted with planning and internal draft critique. ChatGPT/Codex (OpenAI) assisted with planning, literature discovery, mathematical development, code and Lean proof drafting, executable and kernel checks, result interpretation and manuscript revisions. This assistance was material. The reported kernel checks and computational runs support their explicitly stated results; they do not certify

independent human review, novelty or full-manuscript correctness. AI tools are not authors. The human authors remain responsible for verifying all content, citations, analyses and conclusions and for the final scholarly contribution before submission.

References

Alfonsi, A., A. Schied, and A. Slynko (2012). Order book resilience, price manipulation, and the positive portfolio problem. *SIAM Journal on Financial Mathematics* 3. doi:10.1137/110822098.

Alfonsi, A., A. Schied, and F. Klöck (2016). Multivariate transient price impact and matrix-valued positive definite functions. *Mathematics of Operations Research* 41, 914–934.

Almgren, R., and N. Chriss (2001). Optimal execution of portfolio transactions. *Journal of Risk* 3, 5–39.

Amilon, H. (2008). Estimation of an adaptive stock market model with heterogeneous agents. *Journal of Empirical Finance* 15, 342–362.

Andrews, D. W. K., and X. Cheng (2012). Estimation and inference with weak, semi-strong, and strong identification. *Econometrica* 80, 2153–2211.

Banerjee, S., and M. Szydlowski (2026). Trading against algorithms: Price dynamics and risk-sharing in a market with Q-learners. Working paper.

Barberis, N., R. Greenwood, L. Jin, and A. Shleifer (2015). X-CAPM: An extrapolative capital asset pricing model. *Journal of Financial Economics* 115, 1–24.

Barberis, N., R. Greenwood, L. Jin, and A. Shleifer (2018). Extrapolation and bubbles. *Journal of Financial Economics* 129, 203–227.

Barth, D., and R. J. Kahn (2021). Hedge funds and the Treasury cash-futures disconnect. Office of Financial Research Working Paper 21-01.

Batenkov, D., G. Goldman, and Y. Yomdin (2021). Super-resolution of near-colliding point sources. *Information and Inference* 10, 515–572.

Bernstein, S. (1914). Sur la meilleure approximation de $|x|$ par des polynômes de degrés donnés. *Acta Mathematica* 37, 1–57.

Bessembinder, H., W. Maxwell, and K. Venkataraman (2006). Market transparency, liquidity externalities, and institutional trading costs in corporate bonds. *Journal of Financial Economics* 82, 251–288.

Biais, B., T. Foucault, and S. Moinas (2015). Equilibrium fast trading. *Journal of Financial Economics* 116, 292–313.

Boehmer, E., G. Saar, and L. Yu (2005). Lifting the veil: An analysis of pre-trade transparency at the NYSE. *Journal of Finance* 60, 783–815.

Bouchaud, J.-P., Y. Gefen, M. Potters, and M. Wyart (2004). Fluctuations and response in financial

markets: The subtle nature of "random" price changes. *Quantitative Finance* 4, 176–190.

Bouchaud, J.-P., J. Bonart, J. Donier, and M. Gould (2018). *Trades, Quotes and Prices: Financial Markets Under the Microscope*. Cambridge University Press.

Brock, W. A., and C. H. Hommes (1998). Heterogeneous beliefs and routes to chaos in a simple asset pricing model. *Journal of Economic Dynamics and Control* 22, 1235–1274.

Brogaard, J., T. Hendershott, and R. Riordan (2014). High-frequency trading and price discovery. *Review of Financial Studies* 27, 2267–2306.

Brunnermeier, M. K., and L. H. Pedersen (2009). Market liquidity and funding liquidity. *Review of Financial Studies* 22, 2201–2238.

Castelli, V., and T. M. Cover (1996). The relative value of labeled and unlabeled samples in pattern recognition with an unknown mixing parameter. *IEEE Transactions on Information Theory* 42, 2102–2117.

Consolidated Audit Trail, LLC (2026). Consolidated Audit Trail, LLC 2026 Financial and Operating Budget, 31 March 2026 (total expenses $147,044,869). https://www.catnmsplan.com/cat-financial-and-operating-budget.

Cespa, G., and X. Vives (2015). The beauty contest and short-term trading. *Journal of Finance* 70, 2099–2154.

Cochrane, D., and G. H. Orcutt (1949). Application of least squares regression to relationships containing auto-correlated error terms. *Journal of the American Statistical Association* 44, 32–61.

Colliard, J.-E., T. Foucault, and S. Lovo (2022). Algorithmic pricing and liquidity in securities markets. CEPR Discussion Paper 17606.

Crouzet, N., I. Dew-Becker, and C. G. Nathanson (2020). On the effects of restricting short-term investment. *Review of Financial Studies* 33, 1–43.

Dou, W. W., I. Goldstein, and Y. Ji (2025). AI-powered trading, algorithmic collusion, and price efficiency. NBER Working Paper 34054.

Duarte, J., and L. Young (2009). Why is PIN priced? *Journal of Financial Economics* 91, 119–138.

Easley, D., N. M. Kiefer, M. O'Hara, and J. B. Paperman (1996). Liquidity, information, and infrequently traded stocks. *Journal of Finance* 51, 1405–1436.

European Securities and Markets Authority (2016). Guidelines on transaction reporting, order record keeping and clock synchronisation under MiFID II. ESMA/2016/1452.

European Commission (2017). Commission Delegated Regulation (EU) 2017/580 of 24 June 2016 supplementing Regulation (EU) No 600/2014 with regard to regulatory technical standards for the maintenance of relevant data relating to orders in financial instruments (RTS 24), Article 2(1)(b)–(c). *Official Journal of the European Union* L 87, 31.3.2017.

European Commission (2017). Commission Delegated Regulation (EU) 2017/589 of 19 July 2016

supplementing Directive 2014/65/EU with regard to regulatory technical standards specifying the organisational requirements of investment firms engaged in algorithmic trading (RTS 6). *Official Journal of the European Union* L 87, 31.3.2017.

Fan, J. (1991). On the optimal rates of convergence for nonparametric deconvolution problems. *Annals of Statistics* 19, 1257–1272.

Forssell, U., and L. Ljung (1999). Closed-loop identification revisited. *Automatica* 35, 1215–1241.

Franke, R., and F. Westerhoff (2012). Structural stochastic volatility in asset pricing dynamics: Estimation and model contest. *Journal of Economic Dynamics and Control* 36, 1193–1211.

Gatheral, J. (2010). No-dynamic-arbitrage and market impact. *Quantitative Finance* 10, 749–759.

Glosten, L. R., and P. R. Milgrom (1985). Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics* 14, 71–100.

Hasbrouck, J. (1991). Measuring the information content of stock trades. *Journal of Finance* 46, 179–207.

Heinrich, P., and J. Kahn (2018). Strong identifiability and optimal minimax rates for finite mixture estimation. *Annals of Statistics* 46, 2844–2870.

Hendershott, T., C. M. Jones, and A. J. Menkveld (2011). Does algorithmic trading improve liquidity? *Journal of Finance* 66, 1–33.

Hua, Y., and T. K. Sarkar (1990). Matrix pencil method for estimating parameters of exponentially damped/undamped sinusoids in noise. *IEEE Transactions on Acoustics, Speech, and Signal Processing* 38, 814–824.

Katz, R., N. Diab, and D. Batenkov (2023). On the accuracy of Prony's method for recovery of exponential sums with closely spaced exponents. arXiv preprint arXiv:2302.05883.

Kirilenko, A., A. S. Kyle, M. Samadi, and T. Tuzun (2017). The Flash Crash: High-frequency trading in an electronic market. *Journal of Finance* 72, 967–998.

Kong, W., and G. Valiant (2017). Spectrum estimation from samples. *Annals of Statistics* 45, 2218–2247.

Kreiss, H.-O. (1962). Über die Stabilitätsdefinition für Differenzengleichungen die partielle Differentialgleichungen approximieren. *BIT Numerical Mathematics* 2, 153–181.

Kukacka, J., and J. Barunik (2017). Estimation of financial agent-based models with simulated maximum likelihood. *Journal of Economic Dynamics and Control* 85, 21–45.

Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica* 53, 1315–1335.

Le Cam, L. (1986). *Asymptotic Methods in Statistical Decision Theory*. Springer.

Leach, J. C., and A. N. Madhavan (1993). Price experimentation and security market structure. *Review of Financial Studies* 6, 375–404.

Liu, P., and H. Ammari (2022). Super-resolution of positive near-colliding point sources. arXiv preprint arXiv:2212.00536.

Ljung, L. (1999). *System Identification: Theory for the User*, 2nd ed. Prentice Hall, Upper Saddle River, NJ.

Makarov, I., and A. Schoar (2020). Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics* 135, 293–319.

Manole, T., and N. Ho (2022). Refined convergence rates for maximum likelihood estimation under finite mixture models. *Proceedings of the 39th International Conference on Machine Learning*, PMLR 162.

Manski, C. F. (2003). *Partial Identification of Probability Distributions*. Springer, New York.

Moallemi, C. C., B. Park, and B. Van Roy (2008). Strategic execution in the presence of an uninformed arbitrageur. arXiv preprint arXiv:0801.3001.

Moitra, A. (2015). Super-resolution, extremal functions and the condition number of Vandermonde matrices. In *Proceedings of the 47th Annual ACM Symposium on Theory of Computing (STOC)*, 821–830.

Moskowitz, T. J., Y. H. Ooi, and L. H. Pedersen (2012). Time series momentum. *Journal of Financial Economics* 104, 228–250.

Niroula, P., S. Alkhushayni, Y. Chen, M. Bhavsar, and P. Niroula (2026). From human markets to machine markets: Transient amplification, an exact depth condition, and a maximal pre-trade guard. Working paper, Minnesota State University, Mankato.

Obizhaeva, A. A., and J. Wang (2013). Optimal trading strategy and supply/demand dynamics. *Journal of Financial Markets* 16, 1–32.

Pagano, M., and A. Röell (1996). Transparency and liquidity: A comparison of auction and dealer markets with informed trading. *Journal of Finance* 51, 579–611.

Park, B., and B. Van Roy (2015). Adaptive execution: Exploration and learning of price impact. *Operations Research*, doi:10.1287/opre.2015.1415.

Rebafka, T., and F. Roueff (2015). Nonparametric estimation of the mixing density using polynomials. arXiv preprint arXiv:1002.4516.

Rivlin, T. J. (1981). *An Introduction to the Approximation of Functions*. Dover.

Roll, R. (1984). A simple implicit measure of the effective bid-ask spread in an efficient market. *Journal of Finance* 39, 1127–1139.

Roy, R., and T. Kailath (1989). ESPRIT—estimation of signal parameters via rotational invariance techniques. *IEEE Transactions on Acoustics, Speech, and Signal Processing* 37, 984–995.

U.S. Securities and Exchange Commission (2010). Risk management controls for brokers or dealers with market access. Release No. 34-63241.

U.S. Securities and Exchange Commission (2012). Consolidated Audit Trail; Final Rule (Release No. 34-67457, July 18, 2012, 77 FR 45722 at 45790–45795, pp. 220–238; proposing release No. 34-62174, May 26, 2010, 75 FR 32556 at 32598, 32600, pp. 147, 153).

U.S. Securities and Exchange Commission (2016). Joint Industry Plan; Order Approving the National Market System Plan Governing the Consolidated Audit Trail. Release No. 34-79318, November 15, 2016, 81 FR 84696 at 84801, p. 363.

U.S. Securities and Exchange Commission (2024). Notice of Filing of Proposed Rule Change Relating to the 2025 CAT Budget. Release No. 34-101901, December 12, 2024, p. 4 (revised May 2025).

U.S. Securities and Exchange Commission (2025). Order granting exemptive relief from certain requirements of the National Market System Plan governing the Consolidated Audit Trail. Release No. 34-102386, February 10, 2025.

Shahram, M., and P. Milanfar (2005). On the resolvability of sinusoids with nearby frequencies in the presence of noise. *IEEE Transactions on Signal Processing* 53, 2579–2588.

Shleifer, A., and R. W. Vishny (1992). Liquidation values and debt capacity: A market equilibrium approach. *Journal of Finance* 47, 1343–1366.

Sornette, D., S. Lera, J. Lin, and K. Wu (2023). Non-normal interactions create socio-economic bubbles. *Communications Physics* 6, 261.

Staiger, D., and J. H. Stock (1997). Instrumental variables regression with weak instruments. *Econometrica* 65, 557–586.

Trebbi, F., M. B. Zhang, and M. Simkovic (2024). The cost of regulatory compliance in the United States. Working paper.

Trefethen, L. N., and M. Embree (2005). *Spectra and Pseudospectra: The Behavior of Nonnormal Matrices and Operators*. Princeton University Press.

Tremacoldi-Rossi, P. (2025). Manipulating algorithmic markets. Working paper.

Tsybakov, A. B. (2009). *Introduction to Nonparametric Estimation*. Springer.

van Waarde, H. J., J. Eising, H. L. Trentelman, and M. K. Camlibel (2020). Data informativity: A new perspective on data-driven analysis and control. *IEEE Transactions on Automatic Control* 65, 4753–4768.

Varga, R. S., and A. J. Carpenter (1985). On the Bernstein conjecture in approximation theory. *Constructive Approximation* 1, 333–348.

Wagenmaker, A., and K. Jamieson (2020). Active learning for identification of linear dynamical systems. In *Proceedings of the 33rd Conference on Learning Theory*, PMLR 125, 3487–3582.

Wagenmaker, A. J., M. Simchowitz, and K. Jamieson (2021). Task-optimal exploration in linear dynamical systems. In *Proceedings of the 38th International Conference on Machine Learning*, PMLR 139, 10641–10652.

Wellman, M. P. (2006). Methods for empirical game-theoretic analysis. In *Proceedings of the 21st National Conference on Artificial Intelligence (AAAI-06)*, 1552–1555.

Whittle, P. (1953). Estimation and information in stationary time series. *Arkiv för Matematik* 2,

423–434.

Wu, Y., and P. Yang (2020). Optimal estimation of Gaussian mixtures via denoised method of moments. *Annals of Statistics* 48, 1981–2007.